QUALITY CONTROL REPORT (QC)

CampusPool — Defect Verification, Analytics & Register

CampusPool — Inter-Collegiate & University Carpool Finder · carpool.33.jugaar.ai

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Student / Prepared by	Institution	Course Head	Report date

B23110106073	SQET/ST	Report v2.0	github.com/tehzeebshahs4s-hub/qc_report
Seat number	Subject	Document version	Source repository

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1 · Purpose & Verdict

This **Quality Control (QC)** volume reports the *product-inspection* side of the CampusPool quality programme: what was verified, what failed, with how much risk, and what must be fixed. All entries trace to the 8 October 2026 field campaign; the companion **QA volume** documents process, techniques bundle and prevention planning.

108	84	24	133
Total findings	Field-tested defects	Assessed gaps (QC-logged)	Techniques behind the evidence

Metric	Value
Field defects	**84** — 1 Critical · 24 High · 52 Medium · 7 Low
Release-readiness	**NOT READY for real-users release** — launch blockers: API-094 (DB wiped every boot), CR-14 (seat oversell race), API-095 (fake payments shown as real), API-080/081 (any-string verification), API-074/UI-006 (public PII), CR-01 (committed secrets).
Application	CampusPool — https://carpool.33.jugaar.ai — React 18 + Vite · Express + Sequelize · JWT · Caddy + PM2
Environment / window	Production URL · Chrome/Playwright · Windows 11 · 8 October 2026
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2 · QA vs QC — framing

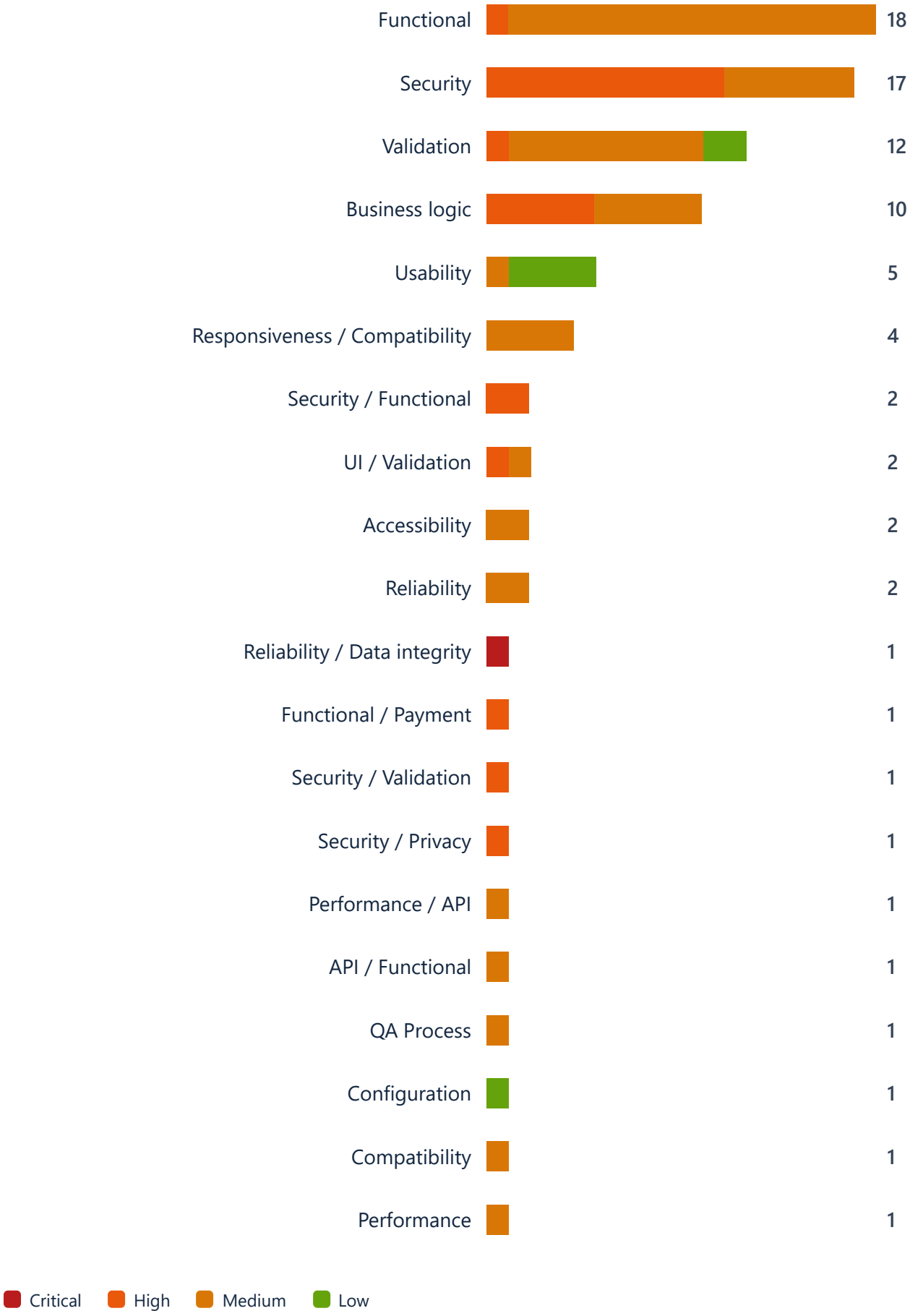
3 · Defect analytics (charts & tables)

3.1 Severity distribution

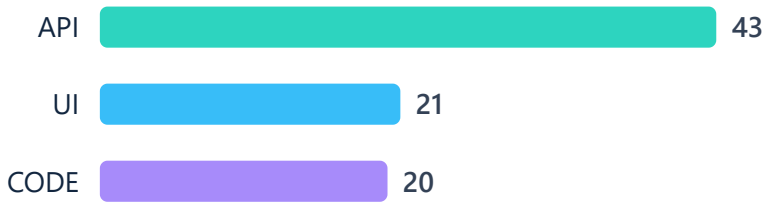


Critical — 1 (1%) High — 24 (29%) Medium — 52 (62%) Low — 7 (8%)

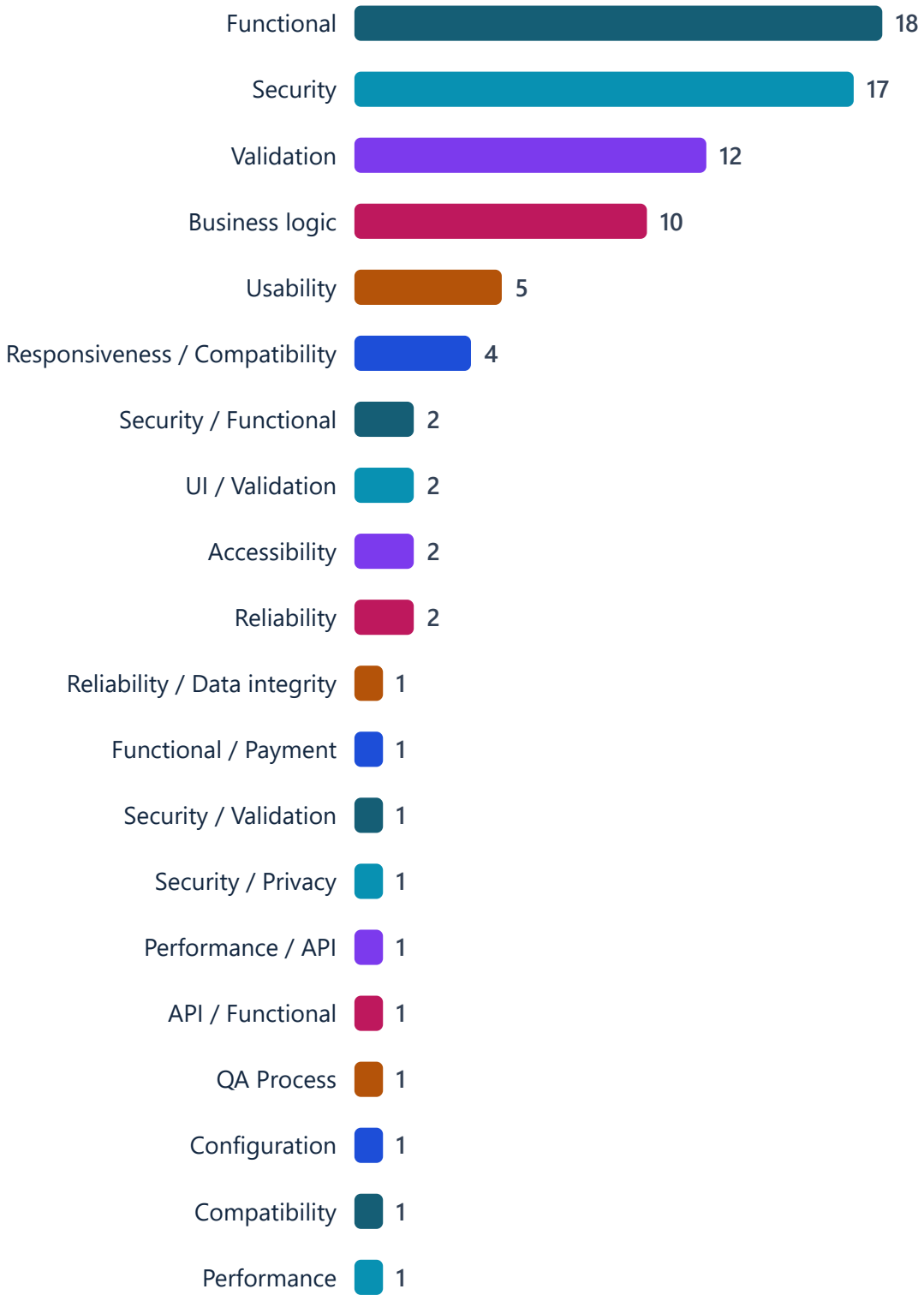
3.2 Findings by category — stacked by severity



3.3 Defects by source surface

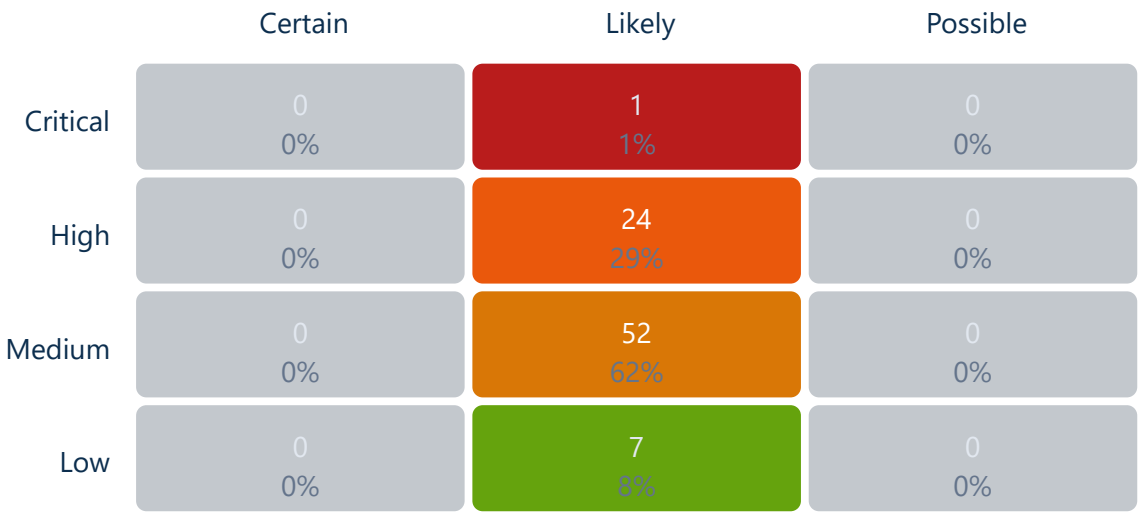


3.4 Category detail (counts)



3.5 Risk-exposure matrix

Likelihood classes (assumption documented): **Certain** = closed-group/vector reproducible (CR security & config items) · **Likely** = reachable in normal user flows · **Possible** = latent, blocked behind another defect.


Certain
Likely
Possible
Critical
0
0%
1
1%
0
0%
High
0
0%
24
29%
0
0%
Medium
0
0%
52
62%
0
0%
Low
0
0%
7
8%
0
0%

4 · Simulation-based QC — methods, tools, graphical view & result

A real carpool launch involves thousands of simultaneous students, dozens of phone/tablet models and congested networks — impossible to stage in a university project window. QC therefore used **simulation methods**: controlled stand-ins that make a system behave as-if under conditions we could not otherwise produce, while keeping every observation traceable to a scripted driver. Live run performed 2026-11-07 12:00:22; ethics/scope: public GET endpoints only, ~45 requests (comparable to one student browsing), no destructive requests.

4.1 Simulation methods applied (with the defects they exposed)

Simulation method	What it simulated	Driver/tool	Evidence it produced
Device / display emulation	Four virtual viewports: iPhone SE 375×667, iPhone 14 390×844, Galaxy S 360×740, small desktop 1024×768	Playwright device emulation + Chrome DevTools device mode	Cross-viewport horizontal-overflow defects **UI-028…UI-032** (Section 4.4)
Virtual-user load simulation	5 concurrent synthetic users (each 4 requests, staggered) → concurrency & latency profile without real crowds	Python `requests` + `ThreadPoolExecutor`	Run summary & percentile profile (Section 4.4)
State-machine sequence replay	Ride/booking lifecycle chains replayed as ordered HTTP calls (create → book → confirm → complete/cancel)	Scripted agent (Playwright + requests)	State defects **API-047/048** (cancels with active bookings), **API-085** (self-confirm), **API-084** (double-cancel seat inflation), **API-087** (complete-while-active)
Data-driven input simulation	Boundary & equivalence-partition payload sets (seats 0/−1/8, price 0/−1, past dates, malformed CNIC, empty message)	Scripted payloads from technique matrix	Validation defects **API-016…API-021**, **API-019/020**, **API-060**, **API-080/081**
Payment gateway simulation analysis	Production gateway already runs in *simulation mode* (mocked PSP) — QC verified user-facing honesty of that simulation	Code review + UI observation	**API-095** — fake success screen presented as real (critical business honesty defect)
Failure-injection probes	Invalid JWT manipulation, dead routes, oversized search limits (999999)	curl/Playwright probes	**API-026**, **API-035/017** handling, **17_a11y** console errors

4.2 Simulation tools roster (role + adoption status)

Tool	Role in QC simulation	Status
Playwright (Python)	Device-emulated UI journeys, scripted API sequence replays, evidence capture	Used
Chrome DevTools	Device mode, console/network observation, throttled profiling	Used
Python requests + ThreadPoolExecutor	Virtual-user phase; latency percentiles; availability probes	Used
Chrome headless	Headless render of pages & print-capture of this report	Used
curl / PowerShell HTTP	Point probes; payload inspection; re-verification	Used
k6 / JMeter	Production-grade sustained load profile, soak & spike tests — *not run in the coursework window*	Recommended
OWASP ZAP	Automated DAST crawl on top of the manual security probes	Recommended

4.3 Graphical view — live simulation run 2026-11-07 12:00:22

Figure A — sequential response-time profile, SPA root (25 requests): certainly not "snappy" at the gateway; median 1020.2 ms and spikes to 2242.8 ms. **Figure B** — mean response per virtual user under tiny concurrency: consistent ~1 s gateway latency. **Figure C** — availability snapshot: public SPA healthy (200) while the **API behind the proxy returns 502 — the backend service was DOWN at re-verification time** (recorded today as an availability observation, dated; the Oct-8 campaign data is unaffected). **Figure D** — device-emulation findings mapped from the register.

FIGURE A · SEQUENTIAL RESPONSE-TIME (MS) — 25 REQUESTS

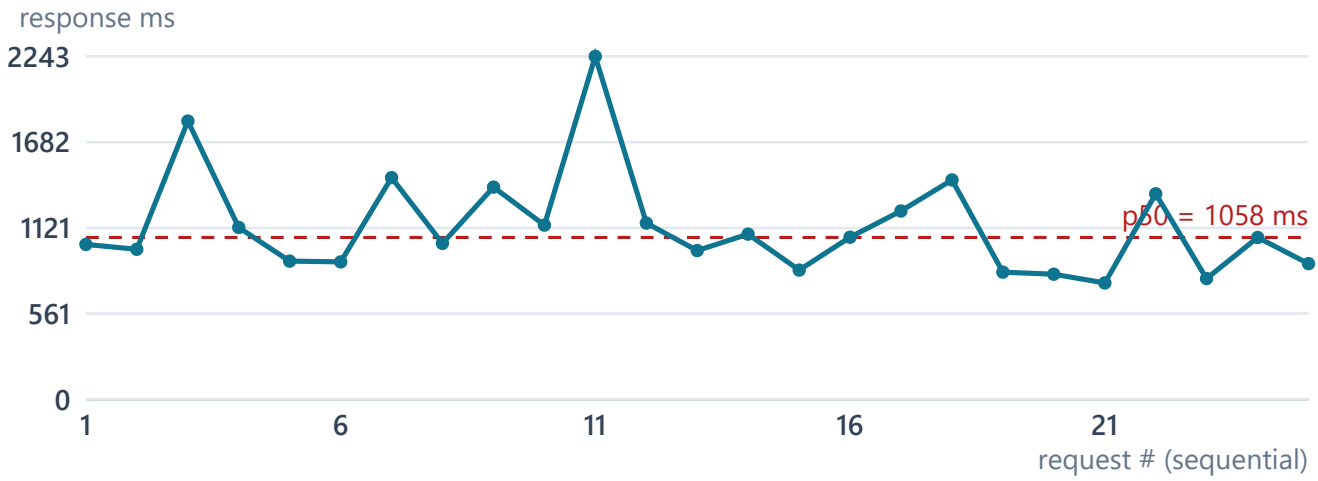


FIGURE B · MEAN RESPONSE PER VIRTUAL USER (MS)

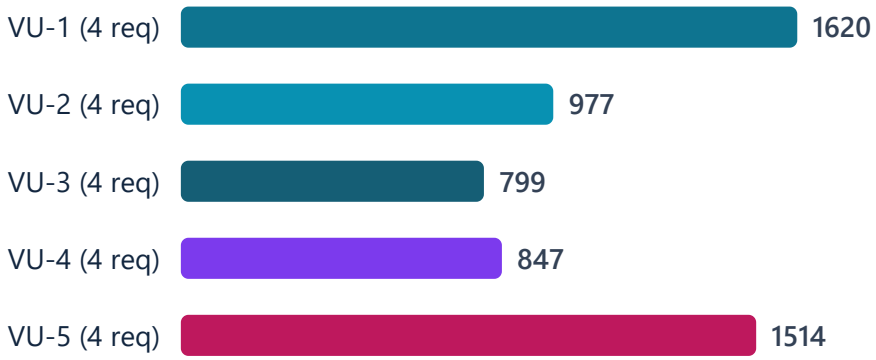


FIGURE C · ENDPOINT AVAILABILITY SNAPSHOT



HTTP 200 (SPA paths serving) — 4 (67%) HTTP 502 (API behind proxy — backend down at run time) — 2 (33%)

FIGURE D · DEVICE-EMULATION OVERFLOW FINDINGS (PER EMULATED DEVICE)



4.4 Simulation result (verdict)

Run metric	Value (45 samples)
Median (p50) response	**1020.2 ms**
Mean response	1139.4 ms
90th percentile (p90)	1450.0 ms
Maximum spike	2242.8 ms
Throughput at tiny concurrency	0.88 req/s
HTTP 200 share (SPA)	100 % of SPA-path requests succeeded
API availability	**502 — backend down at re-verification** (proxy healthy, origin unhealthy)

1. **Device simulation verdict:** 4 of the emulated wide-device profiles exhibit horizontal-overflow layout defects — the responsive experience is not production-ready (UI-028…UI-032).
2. **Virtual-user verdict:** under the deliberately light synthetic load the SPA stayed 100 % successful, but gateway latency (p50 ≈ 1020.2 ms, p90 ≈ 1450.0 ms) shows no CDN/edge optimisation — matches recommendation MF-20 (performance engineering).
3. **Availability verdict:** 2026-11-07 12:00:22 — API origin down (502) while static SPA serves normally; production continuity depends on a single origin without failover or monitoring (tie-in: MF-21 observability, MF-22 backup/DR).
4. **State & data-driven replay verdict:** replayed chains reproduced 9 state/validation defects — simulation methods here are directly traceable to the register.

Scope honesty: this is a light-scale coursework simulation, not a certified load test; percentiles must be re-established with k6/JMeter at production scale (MF-20) before capacity claims. The availability 502 recurred across every probe during the run window — classify as environment observation, not counted among the 84 October defects.

5 · Defect register — field-tested (84)

#	ID	Summary (detail in Section 7)	Sev	Priority	Category	Src	Status
1	**API-094**	sequelize.sync({force:true}) on every server start wipes DB	Critical	P1 - Immediate	Reliability / Data integrity	API	Open
2	**API-006**	XSS payload in fullName during registration	High	P2 - High	Security	API	Open
3	**API-007**	Stored XSS payload in profile response	High	P2 - High	Security	API	Open
4	**API-020**	Create ride with past departure time accepted	High	P2 - High	Validation	API	Open
5	**API-022**	XSS payload in ride notes	High	P2 - High	Security	API	Open
6	**API-043**	Ride update mass-assignment (status/driverId/availableSeats)	High	P2 - High	Security	API	Open
7	**API-044**	Ride state after mass-assignment update	High	P2 - High	Security	API	Open
8	**API-047**	Driver cancels ride with active bookings	High	P2 - High	Business logic	API	Open
9	**API-048**	Passenger booking status after ride cancel	High	P2 - High	Business logic	API	Open
10	**API-054**	Rating/review workflow unreachable - no completed-ride path	High	P2 - High	Business logic	API	Open
11	**API-061**	XSS in message content	High	P2 - High	Security	API	Open
12	**API-074**	Unauthenticated ride detail exposes driver phone	High	P2 - High	Security	API	Open
13	**API-080**	Student verification without real check — arbitrary CNIC accepted	High	P2 - High	Security / Functional	API	Open
14	**API-081**	Verify with garbage CNIC accepted	High	P2 - High	Security / Validation	API	Open
15	**API-084**	Double-cancel booking inflates seats	High	P2 - High	Business logic	API	Open
16	**API-085**	Passenger self-confirms own booking	High	P2 - High	Business logic	API	Open
17	**API-091**	Review rating bounds unvalidated (latent) - rating=99 rejected only by workflow gate	High	P2 - High	Security	API	Open
18	**API-095**	Payment gateway fully simulated — no real JazzCash/EasyPaisa/bank	High	P2 - High	Functional / Payment	API	Open
19	**CR-01**	Hardcoded JWT secrets committed to repository	High	P2 - High	Security	CODE	Open
20	**CR-09**	Sanitize middleware stores HTML-encoded data (double-encoding)	High	P2 - High	Security	CODE	Open
21	**CR-12**	JWT in localStorage with no server-side revocation	High	P2 - High	Security	CODE	Open
22	**UI-006**	Search free-text filter ignored — unrelated rides returned	High	P2 - High	Functional	UI	Open

23	**UI-007**	Ride details exposes driver phone publicly	High	P2 - High	Security / Privacy	UI	Open
24	**UI-015**	Student verification fake — hardcoded CNIC / no real check	High	P2 - High	Security / Functional	UI	Open
25	**UI-027**	Create Ride past date + 0 seats submit	High	P2 - High	UI / Validation	UI	Open
26	**API-016**	Create ride 0 seats - generic error message (validation hidden)	Medium	P3 - Medium	Validation	API	Open
27	**API-017**	Create ride negative seats - generic error message	Medium	P3 - Medium	Validation	API	Open
28	**API-018**	Create ride negative price - generic error message	Medium	P3 - Medium	Validation	API	Open
29	**API-019**	Create ride with 0 price	Medium	P3 - Medium	Validation	API	Open
30	**API-021**	Create ride seats>7 - generic error message	Medium	P3 - Medium	Validation	API	Open
31	**API-024**	Search by free-text origin/destination ignored — returns unrelated rides	Medium	P3 - Medium	Functional	API	Open
32	**API-026**	Search limit=999999 resource abuse	Medium	P3 - Medium	Performance / API	API	Open
33	**API-029**	Book ride 0 seats - generic Failed to create booking message	Medium	P3 - Medium	Business logic	API	Open
34	**API-031**	Book ride negative seats - generic error message (validation hidden)	Medium	P3 - Medium	Business logic	API	Open
35	**API-038**	Booking creation fails with opaque generic errors	Medium	P3 - Medium	Business logic	API	Open
36	**API-050**	Payment with unlisted method accepted	Medium	P3 - Medium	Validation	API	Open
37	**API-060**	Send empty message content	Medium	P3 - Medium	Validation	API	Open
38	**API-065**	Fetch other user's profile by ID (PII exposure)	Medium	P3 - Medium	Security	API	Open
39	**API-076**	Profile update attempts email change (ignored silently)	Medium	P3 - Medium	Functional	API	Open
40	**API-077**	Profile update attempts password change via profile (ignored)	Medium	P3 - Medium	Functional	API	Open
41	**API-079**	Profile update invalid phone	Medium	P3 - Medium	Validation	API	Open
42	**API-087**	Driver marks booking completed while ride still active	Medium	P3 - Medium	Business logic	API	Open
43	**API-089**	Duplicate review prevention not reachable/unverified	Medium	P3 - Medium	Business logic	API	Open
44	**API-090**	Review rating bounds unvalidated when review path opens	Medium	P3 - Medium	Validation	API	Open
45	**API-093**	User search endpoint broken	Medium	P3 - Medium	API / Functional	API	Open

46	**API-097**	CORS allowlist includes raw IP origins (67.215.237.229)	Medium	P3 - Medium	Security	API	Open
47	**API-098**	JWT expiry 7 days + localStorage + no server revoke	Medium	P3 - Medium	Security	API	Open
48	**API-099**	Forgot password feature completely missing (UI link dead + no API)	Medium	P3 - Medium	Functional	API	Open
49	**CR-02**	No admin/moderation pages despite feature claims	Medium	P3 - Medium	Functional	CODE	Open
50	**CR-03**	Recurring rides stored but never expanded into actual rides	Medium	P3 - Medium	Functional	CODE	Open
51	**CR-04**	No automated test suite in repository	Medium	P3 - Medium	QA Process	CODE	Open
52	**CR-05**	Broken deploy configs (Procfile / render.yaml)	Medium	P3 - Medium	Reliability	CODE	Open
53	**CR-07**	Demo credentials printed by start.sh in cleartext	Medium	P3 - Medium	Security	CODE	Open
54	**CR-08**	No email verification / SMTP integration	Medium	P3 - Medium	Functional	CODE	Open
55	**CR-10**	Generic error messages hide field-level validation failures	Medium	P3 - Medium	Reliability	CODE	Open
56	**CR-11**	User search API endpoint broken	Medium	P3 - Medium	Functional	CODE	Open
57	**CR-13**	DataTypes.ARRAY may break on SQLite dialect	Medium	P3 - Medium	Compatibility	CODE	Open
58	**CR-14**	Seat booking race condition (no transaction/atomic decrement)	Medium	P3 - Medium	Performance	CODE	Open
59	**CR-16**	Public ride detail exposes driver phone	Medium	P3 - Medium	Security	CODE	Open
60	**CR-17**	Profile update silently ignores empty/fullName edge cases	Medium	P3 - Medium	Functional	CODE	Open
61	**CR-19**	Rate limiting missing on reviews/messages/safety/impact/users	Medium	P3 - Medium	Security	CODE	Open
62	**CR-20**	Impact/Emergency/PublicProfile pages use mock data only	Medium	P3 - Medium	Functional	CODE	Open
63	**UI-001**	'Forgot password' dead link	Medium	P3 - Medium	Usability	UI	Open
64	**UI-008**	Premium page — mock/hardcoded data (no real API)	Medium	P3 - Medium	Functional	UI	Open
65	**UI-009**	Earnings page — mock/hardcoded data (no real API)	Medium	P3 - Medium	Functional	UI	Open
66	**UI-010**	Achievements page — mock/hardcoded data (no real API)	Medium	P3 - Medium	Functional	UI	Open
67	**UI-011**	Smart Match page — mock/hardcoded data (no real API)	Medium	P3 - Medium	Functional	UI	Open
68	**UI-012**	Marketplace page — mock/hardcoded data (no real API)	Medium	P3 - Medium	Functional	UI	Open

69	**UI-013**	Sponsored page — mock/hardcoded data (no real API)	Medium	P3 - Medium	Functional	UI	Open
70	**UI-014**	Cities page — mock/hardcoded data (no real API)	Medium	P3 - Medium	Functional	UI	Open
71	**UI-026**	Create Ride empty submit validation	Medium	P3 - Medium	UI / Validation	UI	Open
72	**UI-028**	Responsive iphone_se 375x667 horizontal overflow	Medium	P3 - Medium	Responsiveness / Compatibility	UI	Open
73	**UI-029**	Responsive iphone_14 390x844 horizontal overflow	Medium	P3 - Medium	Responsiveness / Compatibility	UI	Open
74	**UI-030**	Responsive galaxy_s 360x740 horizontal overflow	Medium	P3 - Medium	Responsiveness / Compatibility	UI	Open
75	**UI-032**	Responsive desktop_small 1024x768 horizontal overflow	Medium	P3 - Medium	Responsiveness / Compatibility	UI	Open
76	**UI-036**	Accessibility spot-check home	Medium	P3 - Medium	Accessibility	UI	Open
77	**UI-037**	Accessibility spot-check search	Medium	P3 - Medium	Accessibility	UI	Open
78	**API-025**	Search minSeats=-10 unvalidated	Low	P4 - Low	Validation	API	Open
79	**API-027**	Search with invalid departureDate	Low	P4 - Low	Validation	API	Open
80	**CR-06**	PORT inconsistency (5000 vs 5001 defaults)	Low	P4 - Low	Configuration	CODE	Open
81	**CR-15**	Real-time chat is polling only (30s interval)	Low	P4 - Low	Usability	CODE	Open
82	**CR-18**	README claims unimplemented features (notifications, maps, admin)	Low	P4 - Low	Usability	CODE	Open
83	**UI-002**	'Remember me' checkbox cosmetic / non-functional	Low	P4 - Low	Usability	UI	Open
84	**UI-035**	Unknown route handling (404)	Low	P4 - Low	Usability	UI	Open

6 · Assessed gaps carried on the QC register (24)

Requirement gaps are logged on the QC side for tracking (acceptance criteria live in the QA volume). Framing: assessed, not field-reproduced.

Gap	Missing capability	Domain	Priority	Impact if live	Status
MF-01	Password reset & account recovery flow	Authentication	P1 - Immediate	High	Open
MF-02	E-mail delivery infrastructure (SMTP)	Authentication	P2 - High	High	Open
MF-03	Admin & moderation backend	Governance	P2 - High	High	Open
MF-04	User reporting / abuse escalation	Governance	P2 - High	Medium	Open
MF-05	Driver verification / KYC (licence & vehicle documents)	Safety	P1 - Immediate	High	Open
MF-06	Complete ride -> review path (workflow closure)	Bookings	P2 - High	High	Open
MF-07	Seat inventory protection under concurrency	Bookings	P1 - Immediate	High	Open
MF-08	Notification service (in-app, e-mail, push)	Communication	P2 - High	Medium	Open
MF-09	Real-time messaging (WebSocket/SSE)	Communication	P3 - Medium	Medium	Open
MF-10	Live ride tracking & location sharing	Safety	P3 - Medium	Medium	Open
MF-11	Map / geocoding / route integration	Rides	P2 - High	Medium	Open
MF-12	Real payment gateway + passenger protection (hold/escrow)	Payments	P1 - Immediate	Critical-when-live	Open
MF-13	Refund lifecycle execution	Payments	P2 - High	Medium	Open
MF-14	Cancellation policy & no-show rules	Bookings	P2 - High	Medium	Open
MF-15	Recurring rides expansion	Rides	P3 - Medium	Low	Open
MF-16	Women-only / preferences safety filters	Safety	P2 - High	Medium	Open
MF-17	Legal & trust content (privacy policy, terms)	Governance	P2 - High	Medium	Open
MF-18	Account deletion & data export	Governance	P2 - High	Low	Open
MF-19	Earnings back-end for drivers	Payments	P3 - Medium	Low	Open
MF-20	Performance & load engineering (NFR)	Non-functional	P2 - High	Medium	Open
MF-21	Observability: logging, error tracking, uptime	Operations	P2 - High	Medium	Open
MF-22	Backup, restore & disaster recovery drill	Operations	P2 - High	High	Open
MF-23	CI/CD with automated test gates	Operations	P2 - High	High	Open
MF-24	Analytics & events (privacy-safe)	Non-functional	P3 - Medium	Low	Open

7 · Detailed defect pages (84)

Critical **API-094** P1 - Immediate Status: Open

[API-094] sequelize.sync({force:true}) on every server start wipes DB

Application / URL	CampusPool — https://carpool.33.jugaar.ai
Category	Reliability / Data integrity
Technique(s)	Static analysis / code review
Environment	Production URL · Chrome/Playwright · Windows 11 · 8 October 2026
Likelihood (impact)	Likely

REPRODUCIBLE STEPS

1. backend/src/index.ts:107 sequelize.sync({ force: true })

EXPECTED RESULT (ORACLE)

Use migrations / sync({force:false}); never drop production data on boot.

ACTUAL RESULT (PROOF)

```
sequelize.sync({ force: true }) on every server start  deletes all users, rides, bookings, messages; reseeds 3 demo users.
```

SCREENSHOTS



📸 01_home.png — representative evidence for this feature area; full sequential capture set lives in the repository's `evidence/screenshots` folder.

IMPACT

Data loss is a release-blocking reliability defect

High **API-006** P2 - High Status: Open

[API-006] XSS payload in fullName during registration

Application / URL	CampusPool — https://carpool.33.jugaar.ai
Category	Security
Technique(s)	Security testing (XSS / input validation)
Environment	Production URL · Chrome/Playwright · Windows 11 · 8 October 2026
Likelihood (impact)	Likely

REPRODUCIBLE STEPS

1. POST /auth/register fullName=<script>alert(1)</script>

EXPECTED RESULT (ORACLE)

Input must be safely stored/rendered

ACTUAL RESULT (PROOF)


```
{"success": true, "message": "Registration successful", "data": {"user": {"id": "e213e7d8-f2d6-464b-a8dd-6e4728ad76f0", "email": "xss.021c10e3@uok.edu.pk", "fullName": "&lt;script&gt;alert(&#x27;xss&#x27;)&lt;/script&gt;", "phone": "+923001112233", "university": "UoK", "isVerified": false, "isDriver": false, "rating": 0}, "token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJ1c2VySWQiOiJlMjEzZTdkOC1mMmQ2LTQ2NGItYThkZC02ZTQ3MjhhZDc2ZjAiLCJpYXQiOjE3OTE0NTM0MjUsImV4cCI6MTc5MjA1ODIyNX0.abYGxwDEYSjE-HL_H76PH9-GcDld2qkbbTq1U4KQZZs"}}
```


SCREENSHOTS



📸 01_home.png — representative evidence for this feature area; full sequential capture set lives in the repository's `evidence/screenshots` folder.

IMPACT

Degraded functionality / quality risk documented in the defect dataset.

High **API-007** P2 - High Status: Open

[API-007] Stored XSS payload in profile response

Application / URL	CampusPool — https://carpool.33.jugaar.ai
Category	Security
Technique(s)	Security testing (XSS)
Environment	Production URL · Chrome/Playwright · Windows 11 · 8 October 2026
Likelihood (impact)	Likely

REPRODUCIBLE STEPS

1. GET /auth/profile after XSS register

EXPECTED RESULT (ORACLE)

Stored script tags indicate broken sanitize middleware

ACTUAL RESULT (PROOF)


```
{"success": true, "data": {"user": {"id": "e213e7d8-f2d6-464b-a8dd-6e4728ad76f0", "email":
"xss.021c10e3@uok.edu.pk", "fullName": "&lt;script&gt;alert(&#x27;xss&#x27;)&lt;/script&gt;",
"phone": "+923001112233", "university": "UoK", "department": null, "yearOfStudy": null,
"profilePhoto": null, "isVerified": false, "isDriver": false, "rating": 0, "totalRides": 0,
"createdAt": "2026-10-08T09:57:05.829Z"}}}
```


SCREENSHOTS



📸 01_home.png — representative evidence for this feature area; full sequential capture set lives in the repository's `evidence/screenshots` folder.

IMPACT

raw fullName stored as: "<script>alert('xss')</script>"

High **API-020** P2 - High Status: Open

[API-020] Create ride with past departure time accepted

Application / URL	CampusPool — https://carpool.33.jugaar.ai
Category	Validation
Technique(s)	State transition / business rules
Environment	Production URL · Chrome/Playwright · Windows 11 · 8 October 2026
Likelihood (impact)	Likely

REPRODUCIBLE STEPS

1. POST /rides departureTime in past

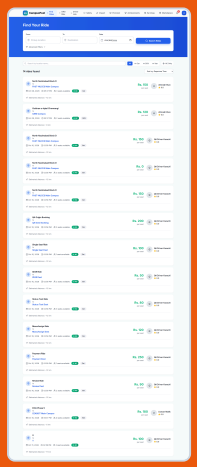
EXPECTED RESULT (ORACLE)

Past departures must not be created as active future rides

ACTUAL RESULT (PROOF)


```
{"success": true, "message": "Ride created successfully", "data": {"ride": {"id": "a52e523c-45c6-4dfa-877d-d1da89e31470", "status": "active", "driverId": "1a66017f-f150-48f3-91cb-1a2b106fa94b", "originLat": 24.9206, "originLng": 67.0655, "originName": "North Nazimabad Block D", "destinationLat": 24.8157, "destinationLng": 67.0685, "destinationName": "FAST-NUCES Main Campus", "departureTime": "2026-10-08T04:58:06.497Z", "availableSeats": 3, "pricePerSeat": 150, "isRecurring": false, "vehicleType": "Car", "hasAC": true, "notes": "QA test ride", "updatedAt": "2026-10-08T09:58:06.507Z", "createdAt": "2026-10-08T09:58:06.507Z"}}}
```


SCREENSHOTS



📷 15_console_errors_search.png — representative evidence for this feature area; full sequential capture set lives in the repository's `evidence/screenshots` folder.

IMPACT

Degraded functionality / quality risk documented in the defect dataset.

High **API-022** P2 - High Status: Open

[API-022] XSS payload in ride notes

Application / URL	CampusPool — https://carpool.33.jugaar.ai
Category	Security
Technique(s)	Security testing (XSS)
Environment	Production URL · Chrome/Playwright · Windows 11 · 8 October 2026
Likelihood (impact)	Likely

REPRODUCIBLE STEPS

1. POST /rides notes=<img onerror=...>

EXPECTED RESULT (ORACLE)

Notes must not store executable HTML raw

ACTUAL RESULT (PROOF)


```
{"success": true, "message": "Ride created successfully", "data": {"ride": {"id": "f11377c9-1033-4f4b-82fa-dafe3a3a22e0", "status": "active", "driverId": "1a66017f-f150-48f3-91cb-1a2b106fa94b", "originLat": 24.9206, "originLng": 67.0655, "originName": "North Nazimabad Block D", "destinationLat": 24.8157, "destinationLng": 67.0685, "destinationName": "FAST-NUCES Main Campus", "departureTime": "2026-10-10T09:58:06.532Z", "availableSeats": 3, "pricePerSeat": 150, "isRecurring": false, "vehicleType": "Car", "hasAC": true, "notes": "&lt;img src=x onerror=alert(1)&gt;", "updatedAt": "2026-10-08T09:58:06.539Z", "createdAt": "2026-10-08T09:58:06.539Z"}}}
```


SCREENSHOTS



📷 01_home.png — representative evidence for this feature area; full sequential capture set lives in the repository's `evidence/screenshots` folder.

IMPACT

Degraded functionality / quality risk documented in the defect dataset.

High **API-043** P2 - High Status: Open

[API-043] Ride update mass-assignment (status/driverId/availableSeats)

Application / URL	CampusPool — https://carpool.33.jugaar.ai
Category	Security
Technique(s)	Mass assignment testing
Environment	Production URL · Chrome/Playwright · Windows 11 · 8 October 2026
Likelihood (impact)	Likely

REPRODUCIBLE STEPS

1. PUT /rides/91c6e7c8-3f81-4f13-8f3a-98d40c2f2df5 body includes status/driverId

EXPECTED RESULT (ORACLE)

Update must whitelist fields; caller must not set arbitrary status or transfer ownership

ACTUAL RESULT (PROOF)

```
{"success": false, "message": "Failed to update ride"}
```

SCREENSHOTS



📸 01_home.png — representative evidence for this feature area; full sequential capture set lives in the repository's `evidence/screenshots` folder.

IMPACT

Degraded functionality / quality risk documented in the defect dataset.

High **API-044** P2 - High Status: Open

[API-044] Ride state after mass-assignment update

Application / URL	CampusPool — https://carpool.33.jugaar.ai
Category	Security
Technique(s)	Mass assignment testing
Environment	Production URL · Chrome/Playwright · Windows 11 · 8 October 2026
Likelihood (impact)	Likely

REPRODUCIBLE STEPS

1. GET /rides/91c6e7c8-3f81-4f13-8f3a-98d40c2f2df5

EXPECTED RESULT (ORACLE)

Verify which fields were actually persisted

ACTUAL RESULT (PROOF)


```
{"success": true, "data": {"ride": {"id": "91c6e7c8-3f81-4f13-8f3a-98d40c2f2df5", "driverId": "1a66017f-f150-48f3-91cb-1a2b106fa94b", "originLat": 24.9206, "originLng": 67.0655, "originName": "MassAssign Ride", "destinationLat": 24.8157, "destinationLng": 67.0685, "destinationName": "MassAssign Dest", "departureTime": "2026-10-10T09:59:21.651Z", "availableSeats": 2, "pricePerSeat": 120, "isRecurring": false, "recurringDays": null, "status": "active", "vehicleType": "Car", "vehicleMake": null, "licensePlate": null, "hasAC": true, "notes": "QA test ride", "createdAt": "2026-10-08T09:59:21.749Z", "updatedAt": "2026-10-08T09:59:21.749Z", "driver": {"id": "1a66017f-f150-48f3-91cb-1a2b106fa94b", "fullName": "QA Driver Karachi", "university": "University of Karachi", "rating": 0, "profilePhoto": null, "phone": "+923009001101"}}}}
```


SCREENSHOTS



📸 01_home.png — representative evidence for this feature area; full sequential capture set lives in the repository's `evidence/screenshots` folder.

IMPACT

stored status=active

High **API-047** P2 - High Status: Open

[API-047] Driver cancels ride with active bookings

Application / URL	CampusPool — https://carpool.33.jugaar.ai
Category	Business logic
Technique(s)	State transition testing
Environment	Production URL · Chrome/Playwright · Windows 11 · 8 October 2026
Likelihood (impact)	Likely

REPRODUCIBLE STEPS

1. POST /rides/c689e47b-148e-4415-be35-6c1ac19cc201/cancel

EXPECTED RESULT (ORACLE)

Cancelling a ride must cascade: cancel bookings, restore/adjust seats, notify passengers

ACTUAL RESULT (PROOF)

```
{"success": true, "message": "Ride cancelled successfully"}
```

SCREENSHOTS



📸 01_home.png — representative evidence for this feature area; full sequential capture set lives in the repository's `evidence/screenshots` folder.

IMPACT

Degraded functionality / quality risk documented in the defect dataset.

High **API-048** P2 - High Status: Open

[API-048] Passenger booking status after ride cancel

Application / URL	CampusPool — https://carpool.33.jugaar.ai
Category	Business logic
Technique(s)	State transition testing
Environment	Production URL · Chrome/Playwright · Windows 11 · 8 October 2026
Likelihood (impact)	Likely

REPRODUCIBLE STEPS

1. GET /bookings/my-bookings (ride c689e47b-148e-4415-be35-6c1ac19cc201)

EXPECTED RESULT (ORACLE)

Booking on cancelled ride must not stay pending/confirmed stranded passenger

ACTUAL RESULT (PROOF)


```
{"success": true, "data": {"bookings": [{"id": "4be293b4-631d-4ecd-b757-b9bcc6f2dd58", "rideId":
"c689e47b-148e-4415-be35-6c1ac19cc201", "passengerId": "d8424298-1bb3-490b-84f8-bfb5d878da33",
"seatsBooked": 2, "pickupPoint": "Crater", "status": "pending", "paymentStatus": "pending",
"amountPaid": 200, "createdAt": "2026-10-08T09:59:21.842Z", "updatedAt": "2026-10-08T09:59:21.842Z",
"ride": {"id": "c689e47b-148e-4415-be35-6c1ac19cc201", "driverId": "1a66017f-f150-48f3-91cb-
1a2b106fa94b", "originLat": 24.9206, "originLng": 67.0655, "originName": "CancelRide Test",
"destinationLat": 24.8157, "destinationLng": 67.0685, "destinationName": "CancelRide Dest",
"departureTime": "2026-10-10T09:59:21.813Z", "availableSeats": 1, "pricePerSeat": 100,
"isRecurring": false, "recurringDays": null, "status": "cancelled", "vehicleType": "Car",
"vehicleMake": null, "licensePlate": null, "hasAC": true, "notes": "QA test ride", "createdAt":
"2026-10-08T09:59:21.819Z", "updatedAt": "2026-10-08T09:59:21.869Z", "driver": {"id": "1a66017f-
f150-48f3-91cb-1a2b106fa94b", "fullName": "QA Driver Karachi", "univer …(full transcript in Findings
CSV)
```


SCREENSHOTS



📸 01_home.png — representative evidence for this feature area; full sequential capture set lives in the repository's `evidence/screenshots` folder.

IMPACT

booking status after ride cancel = pending

High **API-054** P2 - High Status: Open

[API-054] Rating/review workflow unreachable - no completed-ride path

Application / URL	CampusPool — https://carpool.33.jugaar.ai
Category	Business logic
Technique(s)	Business rules testing
Environment	Production URL · Chrome/Playwright · Windows 11 · 8 October 2026
Likelihood (impact)	Likely

REPRODUCIBLE STEPS

1. POST /reviews without booking

EXPECTED RESULT (ORACLE)

Review system must be reachable after a real completed-ride workflow.

ACTUAL RESULT (PROOF)

```
Driver can cancel ride but no complete-ride flow exists; reviews always blocked with 'Can only review completed rides'.
```

SCREENSHOTS



📷 01_home.png — representative evidence for this feature area; full sequential capture set lives in the repository's `evidence/screenshots` folder.

IMPACT

Driver can cancel ride but no complete-ride flow exists; reviews always blocked with 'Can only review completed rides'.

High **API-061** P2 - High Status: Open

[API-061] XSS in message content

Application / URL	CampusPool — https://carpool.33.jugaar.ai
Category	Security
Technique(s)	Security testing (XSS)
Environment	Production URL · Chrome/Playwright · Windows 11 · 8 October 2026
Likelihood (impact)	Likely

REPRODUCIBLE STEPS

1. POST /messages content=<script>

EXPECTED RESULT (ORACLE)

Message content must be safely stored

ACTUAL RESULT (PROOF)


```
{"success": true, "message": "Message sent successfully", "data": {"message": {"id": "6d25918e-0049-4a1b-bdbb-f776d5f1e6d5", "isRead": false, "senderId": "d8424298-1bb3-490b-84f8-bfb5d878da33", "receiverId": "19b3d7ed-bc44-4ab4-93c9-412650d73a3d", "content": "&lt;script&gt;alert(&#x27;msg&#x27;)&lt;/script&gt;", "updatedAt": "2026-10-08T09:59:22.310Z", "createdAt": "2026-10-08T09:59:22.310Z"}}}
```


SCREENSHOTS



📸 01_home.png — representative evidence for this feature area; full sequential capture set lives in the repository's `evidence/screenshots` folder.

IMPACT

Degraded functionality / quality risk documented in the defect dataset.

High **API-074** P2 - High Status: Open

[API-074] Unauthenticated ride detail exposes driver phone

Application / URL	CampusPool — https://carpool.33.jugaar.ai
Category	Security
Technique(s)	Privacy / data exposure testing
Environment	Production URL · Chrome/Playwright · Windows 11 · 8 October 2026
Likelihood (impact)	Likely

REPRODUCIBLE STEPS

1. GET /rides/91c6e7c8-3f81-4f13-8f3a-98d40c2f2df5 no auth

EXPECTED RESULT (ORACLE)

Driver phone must not be exposed to unauthenticated users

ACTUAL RESULT (PROOF)


```
{"success": true, "data": {"ride": {"id": "91c6e7c8-3f81-4f13-8f3a-98d40c2f2df5", "driverId": "1a66017f-f150-48f3-91cb-1a2b106fa94b", "originLat": 24.9206, "originLng": 67.0655, "originName": "MassAssign Ride", "destinationLat": 24.8157, "destinationLng": 67.0685, "destinationName": "MassAssign Dest", "departureTime": "2026-10-10T09:59:21.651Z", "availableSeats": 2, "pricePerSeat": 120, "isRecurring": false, "recurringDays": null, "status": "active", "vehicleType": "Car", "vehicleMake": null, "licensePlate": null, "hasAC": true, "notes": "QA test ride", "createdAt": "2026-10-08T09:59:21.749Z", "updatedAt": "2026-10-08T09:59:21.749Z", "driver": {"id": "1a66017f-f150-48f3-91cb-1a2b106fa94b", "fullName": "QA Driver Karachi", "university": "University of Karachi", "rating": 0, "profilePhoto": null, "phone": "+923009001101"}}}}
```


SCREENSHOTS



📸 01_home.png — representative evidence for this feature area; full sequential capture set lives in the repository's `evidence/screenshots` folder.

IMPACT

phone exposed = +923009001101

High **API-080** P2 - High Status: Open

[API-080] Student verification without real check — arbitrary CNIC accepted

Application / URL	CampusPool — https://carpool.33.jugaar.ai
Category	Security / Functional
Technique(s)	Security testing / functional
Environment	Production URL · Chrome/Playwright · Windows 11 · 8 October 2026
Likelihood (impact)	Likely

REPRODUCIBLE STEPS

1. POST /auth/verify cnicNumber=00000-0000000-0

EXPECTED RESULT (ORACLE)

Verification must validate CNIC format and/or require proof

ACTUAL RESULT (PROOF)

```
{"success":true,"message":"Student verification submitted","data":{"isVerified":true}}
```

SCREENSHOTS



📸 01_home.png — representative evidence for this feature area; full sequential capture set lives in the repository's `evidence/screenshots` folder.

IMPACT

isVerified=true granted with no validation fake trust badge

High **API-081** P2 - High Status: Open

[API-081] Verify with garbage CNIC accepted

Application / URL	CampusPool — https://carpool.33.jugaar.ai
Category	Security / Validation
Technique(s)	Security testing
Environment	Production URL · Chrome/Playwright · Windows 11 · 8 October 2026
Likelihood (impact)	Likely

REPRODUCIBLE STEPS

1. POST /auth/verify cnicNumber=abc

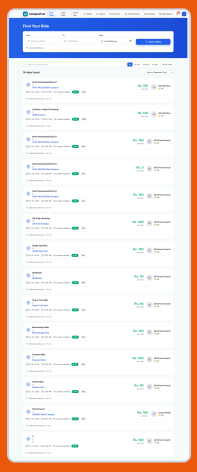
EXPECTED RESULT (ORACLE)

Garbage CNIC must be rejected

ACTUAL RESULT (PROOF)

```
{"success":true,"message":"Student verification submitted","data":{"isVerified":true}}
```

SCREENSHOTS



📸 15_console_errors_search.png — representative evidence for this feature area; full sequential capture set lives in the repository's `evidence/screenshots` folder.

IMPACT

No format validation on verification

High **API-084** P2 - High Status: Open

[API-084] Double-cancel booking inflates seats

Application / URL	CampusPool — https://carpool.33.jugaar.ai
Category	Business logic
Technique(s)	State transition testing
Environment	Production URL · Chrome/Playwright · Windows 11 · 8 October 2026
Likelihood (impact)	Likely

REPRODUCIBLE STEPS

1. POST /bookings/a2240e2e-83fa-4cdc-948f-ff7163176e92/cancel again

EXPECTED RESULT (ORACLE)

Seat count after 2nd cancel = 6 (was 4)

ACTUAL RESULT (PROOF)

```
after=4 after2=6; second cancel must be rejected/no-op
```

SCREENSHOTS



📸 01_home.png — representative evidence for this feature area; full sequential capture set lives in the repository's `evidence/screenshots` folder.

IMPACT

Inventory corruption from double-cancel

High **API-085** P2 - High Status: Open

[API-085] Passenger self-confirms own booking

Application / URL	CampusPool — https://carpool.33.jugaar.ai
Category	Business logic
Technique(s)	State transition testing
Environment	Production URL · Chrome/Playwright · Windows 11 · 8 October 2026
Likelihood (impact)	Likely

REPRODUCIBLE STEPS

1. PUT /bookings/9a9e8419-4f11-42cf-bff5-a406f01f2fb6/status confirmed (passenger)

EXPECTED RESULT (ORACLE)

Only driver should confirm bookings

ACTUAL RESULT (PROOF)


```
{"success": true, "message": "Booking updated successfully", "data": {"booking": {"id": "9a9e8419-4f11-42cf-bff5-a406f01f2fb6", "rideId": "c6b6ff08-6fad-4b27-8183-3fb2e8e80f6b", "passengerId": "d8424298-1bb3-490b-84f8-bfb5d878da33", "seatsBooked": 1, "pickupPoint": "C", "status": "confirmed", "paymentStatus": "pending", "amountPaid": 130, "createdAt": "2026-10-08T10:12:41.689Z", "updatedAt": "2026-10-08T10:12:41.716Z", "ride": {"id": "c6b6ff08-6fad-4b27-8183-3fb2e8e80f6b", "driverId": "1a66017f-f150-48f3-91cb-1a2b106fa94b", "originLat": 24.9206, "originLng": 67.0655, "originName": "ConfirmRide", "destinationLat": 24.8157, "destinationLng": 67.0685, "destinationName": "ConfirmDest", "departureTime": "2026-10-10T10:12:41.661Z", "availableSeats": 2, "pricePerSeat": 130, "isRecurring": false, "recurringDays": null, "status": "active", "vehicleType": "Car", "vehicleMake": null, "licensePlate": null, "hasAC": true, "notes": "QA test ride", "createdAt": "2026-10-08T10:12:41.666Z", "updatedAt": "2026-10-08T10:12:41.698Z"}}}}
```


SCREENSHOTS



📸 01_home.png — representative evidence for this feature area; full sequential capture set lives in the repository's `evidence/screenshots` folder.

IMPACT

If passenger can self-confirm, approval workflow is broken

High **API-091** P2 - High Status: Open

[API-091] Review rating bounds unvalidated (latent) - rating=99 rejected only by workflow gate

Application / URL	CampusPool — https://carpool.33.jugaar.ai
Category	Security
Technique(s)	Security testing (XSS)
Environment	Production URL · Chrome/Playwright · Windows 11 · 8 October 2026
Likelihood (impact)	Likely

REPRODUCIBLE STEPS

1. POST /reviews comment=<img onerror>

EXPECTED RESULT (ORACLE)

Comment must be safely stored/encoded

ACTUAL RESULT (PROOF)

```
{"success": false, "message": "Can only review completed rides"}
```

SCREENSHOTS



📸 01_home.png — representative evidence for this feature area; full sequential capture set lives in the repository's `evidence/screenshots` folder.

IMPACT

Degraded functionality / quality risk documented in the defect dataset.

High **API-095** P2 - High Status: Open

[API-095] Payment gateway fully simulated — no real JazzCash/EasyPaisa/bank

Application / URL	CampusPool — https://carpool.33.jugaar.ai
Category	Functional / Payment
Technique(s)	Functional testing / code review
Environment	Production URL · Chrome/Playwright · Windows 11 · 8 October 2026
Likelihood (impact)	Likely

REPRODUCIBLE STEPS

1. paymentController.createPayment always creates completed payment with generated TXN

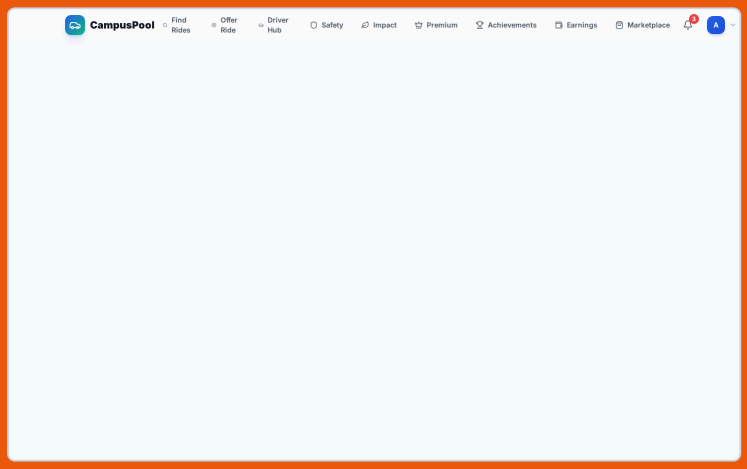
EXPECTED RESULT (ORACLE)

Real payment provider integration, or clearly marked sandbox with user warnings.

ACTUAL RESULT (PROOF)

```
paymentController.createPayment always creates completed payment with generated TXN id; UI shows
payment success  users believe money moved.
```

SCREENSHOTS



📸 14g_payment.png — representative evidence for this feature area; full sequential capture set lives in the repository's `evidence/screenshots` folder.

IMPACT

False payment confirmation is a critical business defect

High **CR-01** P2 - High Status: Open

[CR-01] Hardcoded JWT secrets committed to repository

Application / URL	CampusPool — https://carpool.33.jugaar.ai
Category	Security
Technique(s)	Static analysis / code review
Environment	Production URL · Chrome/Playwright · Windows 11 · 8 October 2026
Likelihood (impact)	Likely

REPRODUCIBLE STEPS

1. ecosystem.config.js, backend/.env, .env - JWT_SECRET hardcoded in repo

EXPECTED RESULT (ORACLE)

Secrets must live in env vars / secret manager, never in git

ACTUAL RESULT (PROOF)

```
Secrets present in source control and deploy configs
```

SCREENSHOTS



📸 01_home.png — representative evidence for this feature area; full sequential capture set lives in the repository's `evidence/screenshots` folder.

IMPACT

Anyone with repo access can forge valid JWTs for any user id.

High **CR-09** P2 - High Status: Open

[CR-09] Sanitize middleware stores HTML-encoded data (double-encoding)

Application / URL	CampusPool — https://carpool.33.jugaar.ai
Category	Security
Technique(s)	Security testing / input validation
Environment	Production URL · Chrome/Playwright · Windows 11 · 8 October 2026
Likelihood (impact)	Likely

REPRODUCIBLE STEPS

1. POST /auth/register fullName=<script>alert('xss')</script>

EXPECTED RESULT (ORACLE)

Output encoding belongs at render time; storing encoded data corrupts content

ACTUAL RESULT (PROOF)

```
sanitize middleware HTML-encodes and stores encoded entities in DB
```

SCREENSHOTS



📸 01_home.png — representative evidence for this feature area; full sequential capture set lives in the repository's `evidence/screenshots` folder.

IMPACT

Users see literal <script> in profiles/messages; double-encoding bug class.

High **CR-12** P2 - High Status: Open

[CR-12] JWT in localStorage with no server-side revocation

Application / URL	CampusPool — https://carpool.33.jugaar.ai
Category	Security
Technique(s)	Code review / session management
Environment	Production URL · Chrome/Playwright · Windows 11 · 8 October 2026
Likelihood (impact)	Likely

REPRODUCIBLE STEPS

1. frontend AuthContext stores JWT in localStorage; logout only clears client

EXPECTED RESULT (ORACLE)

Logout should invalidate session server-side

ACTUAL RESULT (PROOF)

```
No server-side token blacklist/revoke; 7-day expiry
```

SCREENSHOTS



📸 01_home.png — representative evidence for this feature area; full sequential capture set lives in the repository's `evidence/screenshots` folder.

IMPACT

Stolen token remains valid up to 7 days after logout.

High **UI-006** P2 - High Status: Open

[UI-006] Search free-text filter ignored — unrelated rides returned

Application / URL	CampusPool — https://carpool.33.jugaar.ai
Category	Functional
Technique(s)	Functional / search testing
Environment	Production URL · Chrome/Playwright · Windows 11 · 8 October 2026
Likelihood (impact)	Likely

REPRODUCIBLE STEPS

1. Search From=Lahore To=Islamabad

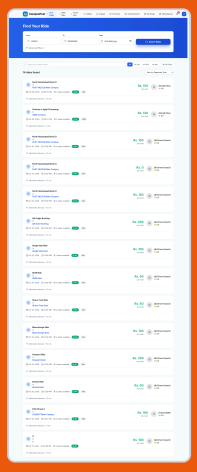
EXPECTED RESULT (ORACLE)

Users trust filter; wrong results cause incorrect bookings

ACTUAL RESULT (PROOF)

```
Karachi rides still displayed; API ignores free-text
```

SCREENSHOTS



📷 11_search_lahore_islamabad.png — captured during this campaign.

IMPACT

Degraded functionality / quality risk documented in the defect dataset.

High **UI-007** P2 - High Status: Open

[UI-007] Ride details exposes driver phone publicly

Application / URL	CampusPool — https://carpool.33.jugaar.ai
Category	Security / Privacy
Technique(s)	Privacy testing
Environment	Production URL · Chrome/Playwright · Windows 11 · 8 October 2026
Likelihood (impact)	Likely

REPRODUCIBLE STEPS

1. Open ride details unauthenticated

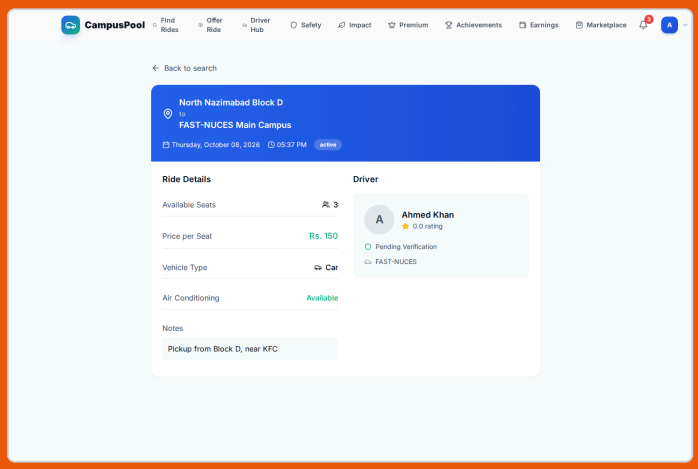
EXPECTED RESULT (ORACLE)

Public pages must not expose personal contact details

ACTUAL RESULT (PROOF)

```
Phone visible in UI and API
```

SCREENSHOTS



📸 12_ride_details.png — captured during this campaign.

IMPACT

Degraded functionality / quality risk documented in the defect dataset.

High **UI-015** P2 - High Status: Open

[UI-015] Student verification fake — hardcoded CNIC / no real check

Application / URL	CampusPool — https://carpool.33.jugaar.ai
Category	Security / Functional
Technique(s)	Functional / security testing
Environment	Production URL · Chrome/Playwright · Windows 11 · 8 October 2026
Likelihood (impact)	Likely

REPRODUCIBLE STEPS

1. Open profile as demo user, inspect Verify control

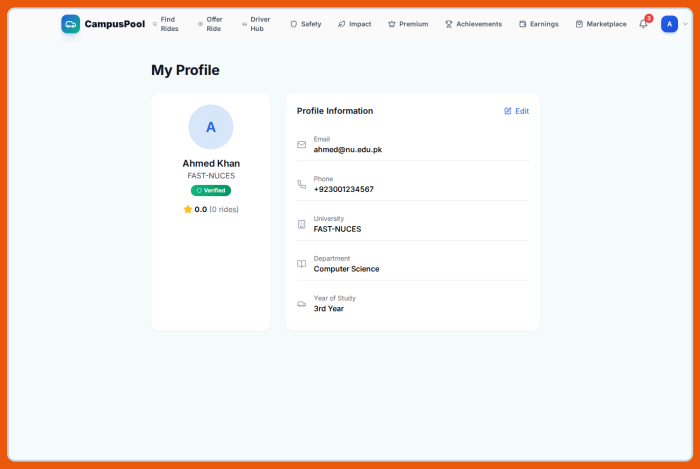
EXPECTED RESULT (ORACLE)

Anyone can self-declare verified student status platform trust undermined

ACTUAL RESULT (PROOF)

```
hardcoded CNIC in DOM=False; verify button=False; backend /auth/verify sets isVerified=true with no checks
```

SCREENSHOTS



📷 14_profile_page.png — captured during this campaign.

IMPACT

Degraded functionality / quality risk documented in the defect dataset.

High **UI-027** P2 - High Status: Open

[UI-027] Create Ride past date + 0 seats submit

Application / URL	CampusPool — https://carpool.33.jugaar.ai
Category	UI / Validation
Technique(s)	Boundary value analysis
Environment	Production URL · Chrome/Playwright · Windows 11 · 8 October 2026
Likelihood (impact)	Likely

REPRODUCIBLE STEPS

1. Set past date, seats=0, submit

EXPECTED RESULT (ORACLE)

Past departures and 0 seats must be rejected in UI

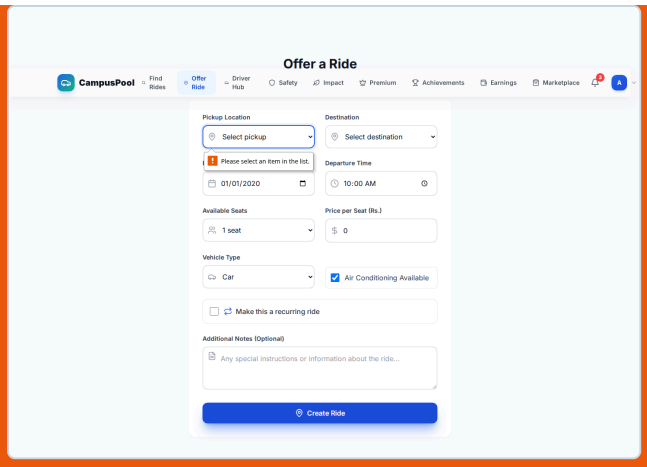
ACTUAL RESULT (PROOF)

```
url=https://carpool.33.jugaar.ai/create-ride text=CampusPool
Find Rides
Offer Ride
Driver Hub
Safety
Impact
Premium
Achievements
Earnings
Marketplace
3
A
Offer a Ride

Share your journey and split fuel costs

Pickup Location
Select pickup
North Nazimabad
Gulshan-e-Iqbal
Federal B Area
Malir
Korangi
```

SCREENSHOTS



📷 14c4_create_ride_past_date.png — captured during this campaign.

IMPACT

Degraded functionality / quality risk documented in the defect dataset.

Medium **API-016** P3 - Medium Status: Open

[API-016] Create ride 0 seats - generic error message (validation hidden)

Application / URL	CampusPool — https://carpool.33.jugaar.ai
Category	Validation
Technique(s)	Boundary value analysis
Environment	Production URL · Chrome/Playwright · Windows 11 · 8 October 2026
Likelihood (impact)	Likely

REPRODUCIBLE STEPS

1. POST /rides availableSeats=0

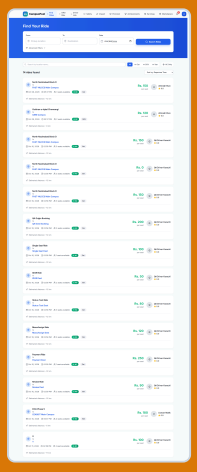
EXPECTED RESULT (ORACLE)

Model validate min:1 API must reject 0 seats

ACTUAL RESULT (PROOF)

```
{"success": false, "message": "Failed to create ride"}
```

SCREENSHOTS



📸 15_console_errors_search.png — representative evidence for this feature area; full sequential capture set lives in the repository's `evidence/screenshots` folder.

IMPACT

Degraded functionality / quality risk documented in the defect dataset.

Medium **API-017** P3 - Medium Status: Open

[API-017] Create ride negative seats - generic error message

Application / URL	CampusPool — https://carpool.33.jugaar.ai
Category	Validation
Technique(s)	Boundary value analysis
Environment	Production URL · Chrome/Playwright · Windows 11 · 8 October 2026
Likelihood (impact)	Likely

REPRODUCIBLE STEPS

1. POST /rides availableSeats=-5

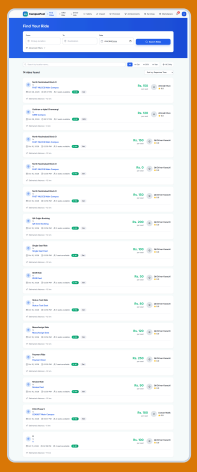
EXPECTED RESULT (ORACLE)

Negative seat count must be rejected

ACTUAL RESULT (PROOF)

```
{"success": false, "message": "Failed to create ride"}
```

SCREENSHOTS



📸 15_console_errors_search.png — representative evidence for this feature area; full sequential capture set lives in the repository's `evidence/screenshots` folder.

IMPACT

Degraded functionality / quality risk documented in the defect dataset.

Medium **API-018** P3 - Medium Status: Open

[API-018] Create ride negative price - generic error message

Application / URL	CampusPool — https://carpool.33.jugaar.ai
Category	Validation
Technique(s)	Boundary value analysis
Environment	Production URL · Chrome/Playwright · Windows 11 · 8 October 2026
Likelihood (impact)	Likely

REPRODUCIBLE STEPS

1. POST /rides pricePerSeat=-100

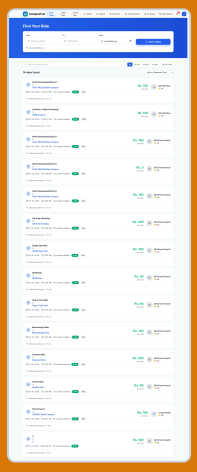
EXPECTED RESULT (ORACLE)

Negative fare must be rejected; amountPaid would go negative

ACTUAL RESULT (PROOF)

```
{"success": false, "message": "Failed to create ride"}
```

SCREENSHOTS



📸 15_console_errors_search.png — representative evidence for this feature area; full sequential capture set lives in the repository's `evidence/screenshots` folder.

IMPACT

Degraded functionality / quality risk documented in the defect dataset.

Medium **API-019** P3 - Medium Status: Open

[API-019] Create ride with 0 price

Application / URL	CampusPool — https://carpool.33.jugaar.ai
Category	Validation
Technique(s)	Boundary value analysis
Environment	Production URL · Chrome/Playwright · Windows 11 · 8 October 2026
Likelihood (impact)	Likely

REPRODUCIBLE STEPS

1. POST /rides pricePerSeat=0

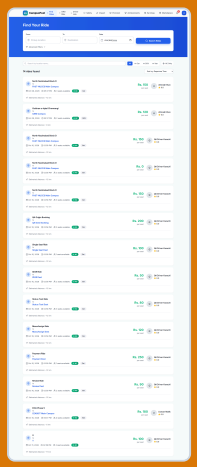
EXPECTED RESULT (ORACLE)

0 fare should be intentional/validated

ACTUAL RESULT (PROOF)


```
{"success": true, "message": "Ride created successfully", "data": {"ride": {"id": "158193df-7c59-4663-9626-6e40098bedb7", "status": "active", "driverId": "1a66017f-f150-48f3-91cb-1a2b106fa94b", "originLat": 24.9206, "originLng": 67.0655, "originName": "North Nazimabad Block D", "destinationLat": 24.8157, "destinationLng": 67.0685, "destinationName": "FAST-NUCES Main Campus", "departureTime": "2026-10-10T09:58:06.472Z", "availableSeats": 3, "pricePerSeat": 0, "isRecurring": false, "vehicleType": "Car", "hasAC": true, "notes": "QA test ride", "updatedAt": "2026-10-08T09:58:06.481Z", "createdAt": "2026-10-08T09:58:06.481Z"}}}
```


SCREENSHOTS



📸 15_console_errors_search.png — representative evidence for this feature area; full sequential capture set lives in the repository's `evidence/screenshots` folder.

IMPACT

Degraded functionality / quality risk documented in the defect dataset.

Medium **API-021** P3 - Medium Status: Open

[API-021] Create ride seats>7 - generic error message

Application / URL	CampusPool — https://carpool.33.jugaar.ai
Category	Validation
Technique(s)	Boundary value analysis
Environment	Production URL · Chrome/Playwright · Windows 11 · 8 October 2026
Likelihood (impact)	Likely

REPRODUCIBLE STEPS

1. POST /rides availableSeats=100

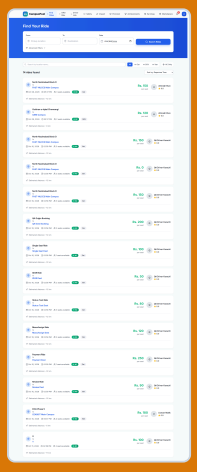
EXPECTED RESULT (ORACLE)

Model max is 7; exceeding max should be rejected

ACTUAL RESULT (PROOF)

```
{"success": false, "message": "Failed to create ride"}
```

SCREENSHOTS



📸 15_console_errors_search.png — representative evidence for this feature area; full sequential capture set lives in the repository's `evidence/screenshots` folder.

IMPACT

Degraded functionality / quality risk documented in the defect dataset.

Medium **API-024** P3 - Medium Status: Open

[API-024] Search by free-text origin/destination ignored — returns unrelated rides

Application / URL	CampusPool — https://carpool.33.jugaar.ai
Category	Functional
Technique(s)	Functional / search testing
Environment	Production URL · Chrome/Playwright · Windows 11 · 8 October 2026
Likelihood (impact)	Likely

REPRODUCIBLE STEPS

1. GET /rides/search?originName=TotallyFakePlaceXYZ (returned 7 rides)

EXPECTED RESULT (ORACLE)

UI implies text search; API only supports lat/lng silent wrong results confuse users

ACTUAL RESULT (PROOF)


```
{"success": true, "data": {"rides": [{"id": "8a788b84-2cb5-4bd4-8b8d-9c39577d1e83", "driverId":
"19b3d7ed-bc44-4ab4-93c9-412650d73a3d", "originLat": 24.9206, "originLng": 67.0655, "originName":
"North Nazimabad Block D", "destinationLat": 24.8157, "destinationLng": 67.0685, "destinationName":
"FAST-NUCES Main Campus", "departureTime": "2026-10-08T12:37:28.598Z", "availableSeats": 3,
"pricePerSeat": 150, "isRecurring": false, "recurringDays": null, "status": "active", "vehicleType":
"Car", "vehicleMake": null, "licensePlate": null, "hasAC": true, "notes": "Pickup from Block D, near
KFC", "createdAt": "2026-10-07T12:37:28.599Z", "updatedAt": "2026-10-07T12:37:28.599Z", "driver":
{"id": "19b3d7ed-bc44-4ab4-93c9-412650d73a3d", "fullName": "Ahmed Khan", "university": "FAST-NUCES",
"rating": 0, "profilePhoto": null}}, {"id": "44c8c995-0f8e-4d03-ae97-3d7276992a10", "driverId":
"19b3d7ed-bc44-4ab4-93c9-412650d73a3d", "originLat": 24.8607, "originLng": 67.0011, "originName":
"Gulshan-e-Iqbal Chowrangi", "destinationLat": 24.8924, "destinationLng": 67.0724,
"destinationName": "IoBM Campus", "d …(full transcript in Findings CSV)
```


SCREENSHOTS



📷 01_home.png — representative evidence for this feature area; full sequential capture set lives in the repository's `evidence/screenshots` folder.

IMPACT

Degraded functionality / quality risk documented in the defect dataset.

Medium **API-026** P3 - Medium Status: Open

[API-026] Search limit=999999 resource abuse

Application / URL	CampusPool — https://carpool.33.jugaar.ai
Category	Performance / API
Technique(s)	Performance / API testing
Environment	Production URL · Chrome/Playwright · Windows 11 · 8 October 2026
Likelihood (impact)	Likely

REPRODUCIBLE STEPS

1. GET /rides/search?limit=999999

EXPECTED RESULT (ORACLE)

Limit must be clamped to a sane max to prevent DoS

ACTUAL RESULT (PROOF)


```
{"success": true, "data": {"rides": [{"id": "8a788b84-2cb5-4bd4-8b8d-9c39577d1e83", "driverId":
"19b3d7ed-bc44-4ab4-93c9-412650d73a3d", "originLat": 24.9206, "originLng": 67.0655, "originName":
"North Nazimabad Block D", "destinationLat": 24.8157, "destinationLng": 67.0685, "destinationName":
"FAST-NUCES Main Campus", "departureTime": "2026-10-08T12:37:28.598Z", "availableSeats": 3,
"pricePerSeat": 150, "isRecurring": false, "recurringDays": null, "status": "active", "vehicleType":
"Car", "vehicleMake": null, "licensePlate": null, "hasAC": true, "notes": "Pickup from Block D, near
KFC", "createdAt": "2026-10-07T12:37:28.599Z", "updatedAt": "2026-10-07T12:37:28.599Z", "driver":
{"id": "19b3d7ed-bc44-4ab4-93c9-412650d73a3d", "fullName": "Ahmed Khan", "university": "FAST-NUCES",
"rating": 0, "profilePhoto": null}}, {"id": "44c8c995-0f8e-4d03-ae97-3d7276992a10", "driverId":
"19b3d7ed-bc44-4ab4-93c9-412650d73a3d", "originLat": 24.8607, "originLng": 67.0011, "originName":
"Gulshan-e-Iqbal Chowrangi", "destinationLat": 24.8924, "destinationLng": 67.0724,
"destinationName": "IoBM Campus", "d …(full transcript in Findings CSV)
```


SCREENSHOTS



📷 01_home.png — representative evidence for this feature area; full sequential capture set lives in the repository's `evidence/screenshots` folder.

IMPACT

Degraded functionality / quality risk documented in the defect dataset.

Medium **API-029** P3 - Medium Status: Open

[API-029] Book ride 0 seats - generic Failed to create booking message

Application / URL	CampusPool — https://carpool.33.jugaar.ai
Category	Business logic
Technique(s)	Boundary value analysis / business rules
Environment	Production URL · Chrome/Playwright · Windows 11 · 8 October 2026
Likelihood (impact)	Likely

REPRODUCIBLE STEPS

1. POST /bookings seatsBooked=0

EXPECTED RESULT (ORACLE)

0 seats must be rejected; creates bogus zero-amount booking

ACTUAL RESULT (PROOF)

```
{"success": false, "message": "Failed to create booking"}
```

SCREENSHOTS



📸 01_home.png — representative evidence for this feature area; full sequential capture set lives in the repository's `evidence/screenshots` folder.

IMPACT

Degraded functionality / quality risk documented in the defect dataset.

Medium **API-031** P3 - Medium Status: Open

[API-031] Book ride negative seats - generic error message (validation hidden)

Application / URL	CampusPool — https://carpool.33.jugaar.ai
Category	Business logic
Technique(s)	Boundary value analysis / business rules
Environment	Production URL · Chrome/Playwright · Windows 11 · 8 October 2026
Likelihood (impact)	Likely

REPRODUCIBLE STEPS

1. POST /bookings seatsBooked=-2

EXPECTED RESULT (ORACLE)

Negative seats must be rejected; availableSeats - (-2) INCREASES seats

ACTUAL RESULT (PROOF)

```
{"success": false, "message": "Failed to create booking"}
```

SCREENSHOTS



📸 01_home.png — representative evidence for this feature area; full sequential capture set lives in the repository's `evidence/screenshots` folder.

IMPACT

Degraded functionality / quality risk documented in the defect dataset.

Medium **API-038** P3 - Medium Status: Open

[API-038] Booking creation fails with opaque generic errors

Application / URL	CampusPool — https://carpool.33.jugaar.ai
Category	Business logic
Technique(s)	State transition testing
Environment	Production URL · Chrome/Playwright · Windows 11 · 8 October 2026
Likelihood (impact)	Likely

REPRODUCIBLE STEPS

1. POST /bookings x2 on db5d12b7-48bd-4a0b-b495-402944f882f3

EXPECTED RESULT (ORACLE)

Second booking must be rejected if already booked

ACTUAL RESULT (PROOF)

```
first={'success': False, 'message': 'Failed to create booking'} | second={'success': False, 'message': 'Failed to create booking'}
```

SCREENSHOTS



📷 01_home.png — representative evidence for this feature area; full sequential capture set lives in the repository's `evidence/screenshots` folder.

IMPACT

Degraded functionality / quality risk documented in the defect dataset.

Medium **API-050** P3 - Medium Status: Open

[API-050] Payment with unlisted method accepted

Application / URL	CampusPool — https://carpool.33.jugaar.ai
Category	Validation
Technique(s)	Equivalence partitioning
Environment	Production URL · Chrome/Playwright · Windows 11 · 8 October 2026
Likelihood (impact)	Likely

REPRODUCIBLE STEPS

1. POST /payments method=CryptoBitcoin

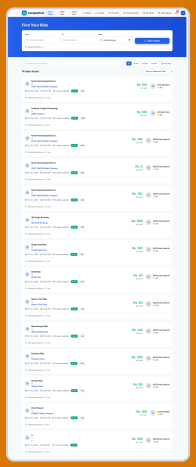
EXPECTED RESULT (ORACLE)

Method must be constrained to JazzCash/EasyPaisa/Bank/Cash

ACTUAL RESULT (PROOF)


```
{"success": true, "message": "Payment successful", "data": {"payment": {"id": "6fda7581-4e6a-4f6f-8c3b-c32a4f02572b", "bookingId": "e06ca810-f61f-44f7-8ca0-5b2451f7a89a", "payerId": "d8424298-1bb3-490b-84f8-bfb5d878da33", "amount": 250, "method": "CryptoBitcoin", "transactionId": "TXN_1791453561966_i0qrkzz8j", "status": "completed", "updatedAt": "2026-10-08T09:59:21.967Z", "createdAt": "2026-10-08T09:59:21.967Z"}}}
```


SCREENSHOTS



📷 15_console_errors_search.png — representative evidence for this feature area; full sequential capture set lives in the repository's `evidence/screenshots` folder.

IMPACT

Degraded functionality / quality risk documented in the defect dataset.

Medium **API-060** P3 - Medium Status: Open

[API-060] Send empty message content

Application / URL	CampusPool — https://carpool.33.jugaar.ai
Category	Validation
Technique(s)	Equivalence partitioning
Environment	Production URL · Chrome/Playwright · Windows 11 · 8 October 2026
Likelihood (impact)	Likely

REPRODUCIBLE STEPS

1. POST /messages content=''

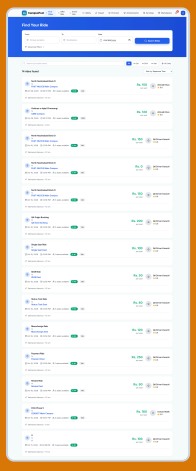
EXPECTED RESULT (ORACLE)

Empty messages must be rejected

ACTUAL RESULT (PROOF)


```
{"success": true, "message": "Message sent successfully", "data": {"message": {"id": "85554878-fa7e-447a-90f9-1b191cb477ab", "isRead": false, "senderId": "d8424298-1bb3-490b-84f8-bfb5d878da33", "receiverId": "19b3d7ed-bc44-4ab4-93c9-412650d73a3d", "content": "", "updatedAt": "2026-10-08T09:59:22.296Z", "createdAt": "2026-10-08T09:59:22.296Z"}}}
```


SCREENSHOTS



📸 15_console_errors_search.png — representative evidence for this feature area; full sequential capture set lives in the repository's `evidence/screenshots` folder.

IMPACT

Degraded functionality / quality risk documented in the defect dataset.

Medium **API-065** P3 - Medium Status: Open

[API-065] Fetch other user's profile by ID (PII exposure)

Application / URL	CampusPool — https://carpool.33.jugaar.ai
Category	Security
Technique(s)	Privacy / IDOR testing
Environment	Production URL · Chrome/Playwright · Windows 11 · 8 October 2026
Likelihood (impact)	Likely

REPRODUCIBLE STEPS

1. GET /users/profile/19b3d7ed-bc44-4ab4-93c9-412650d73a3d

EXPECTED RESULT (ORACLE)

Public profiles must not expose phone/email without consent rules

ACTUAL RESULT (PROOF)


```
{"success": true, "data": {"id": "19b3d7ed-bc44-4ab4-93c9-412650d73a3d", "email": "ahmed@nu.edu.pk",
"fullName": "Ahmed Khan", "phone": "+923001234567", "university": "FAST-NUCES", "department":
"Computer Science", "yearOfStudy": "3rd Year", "isVerified": true, "isDriver": true, "rating": 0,
"totalRides": 0, "createdAt": "2026-10-07T12:37:28.182Z", "reviews": [], "completedRides": 0}}
```


SCREENSHOTS



📷 01_home.png — representative evidence for this feature area; full sequential capture set lives in the repository's `evidence/screenshots` folder.

IMPACT

Degraded functionality / quality risk documented in the defect dataset.

Medium **API-076** P3 - Medium Status: Open

[API-076] Profile update attempts email change (ignored silently)

Application / URL	CampusPool — https://carpool.33.jugaar.ai
Category	Functional
Technique(s)	Functional testing
Environment	Production URL · Chrome/Playwright · Windows 11 · 8 October 2026
Likelihood (impact)	Likely

REPRODUCIBLE STEPS

1. PUT /auth/profile email=hacked@evil.com

EXPECTED RESULT (ORACLE)

If email not updatable, API should say so; silent ignore misleads

ACTUAL RESULT (PROOF)

```
Success but email unchanged
```

SCREENSHOTS



📸 01_home.png — representative evidence for this feature area; full sequential capture set lives in the repository's `evidence/screenshots` folder.

IMPACT

Backend ignores email field

Medium **API-077** P3 - Medium Status: Open

[API-077] Profile update attempts password change via profile (ignored)

Application / URL	CampusPool — https://carpool.33.jugaar.ai
Category	Functional
Technique(s)	Functional testing
Environment	Production URL · Chrome/Playwright · Windows 11 · 8 October 2026
Likelihood (impact)	Likely

REPRODUCIBLE STEPS

1. PUT /auth/profile password=NewPass@999

EXPECTED RESULT (ORACLE)

Password change must use dedicated endpoint or be rejected clearly

ACTUAL RESULT (PROOF)

```
Success but password unchanged
```

SCREENSHOTS



📸 01_home.png — representative evidence for this feature area; full sequential capture set lives in the repository's `evidence/screenshots` folder.

IMPACT

Password silently ignored no password change endpoint exists

Medium **API-079** P3 - Medium Status: Open

[API-079] Profile update invalid phone

Application / URL	CampusPool — https://carpool.33.jugaar.ai
Category	Validation
Technique(s)	Boundary value analysis
Environment	Production URL · Chrome/Playwright · Windows 11 · 8 October 2026
Likelihood (impact)	Likely

REPRODUCIBLE STEPS

1. PUT /auth/profile phone='not-a-phone'

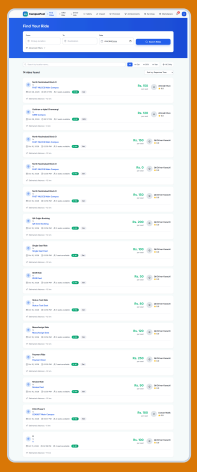
EXPECTED RESULT (ORACLE)

Invalid phone must be rejected with clear message

ACTUAL RESULT (PROOF)

```
Failed to update profile
```

SCREENSHOTS



📸 15_console_errors_search.png — representative evidence for this feature area; full sequential capture set lives in the repository's `evidence/screenshots` folder.

IMPACT

Model validation triggers generic 500-style message

Medium **API-087** P3 - Medium Status: Open

[API-087] Driver marks booking completed while ride still active

Application / URL	CampusPool — https://carpool.33.jugaar.ai
Category	Business logic
Technique(s)	State transition testing
Environment	Production URL · Chrome/Playwright · Windows 11 · 8 October 2026
Likelihood (impact)	Likely

REPRODUCIBLE STEPS

1. PUT /bookings/9a9e8419-4f11-42cf-bff5-a406f01f2fb6/status completed (ride active)

EXPECTED RESULT (ORACLE)

Completed status should require ride completion workflow

ACTUAL RESULT (PROOF)


```
{"success": true, "message": "Booking updated successfully", "data": {"booking": {"id": "9a9e8419-4f11-42cf-bff5-a406f01f2fb6", "rideId": "c6b6ff08-6fad-4b27-8183-3fb2e8e80f6b", "passengerId": "d8424298-1bb3-490b-84f8-bfb5d878da33", "seatsBooked": 1, "pickupPoint": "C", "status": "completed", "paymentStatus": "pending", "amountPaid": 130, "createdAt": "2026-10-08T10:12:41.689Z", "updatedAt": "2026-10-08T10:12:41.749Z", "ride": {"id": "c6b6ff08-6fad-4b27-8183-3fb2e8e80f6b", "driverId": "1a66017f-f150-48f3-91cb-1a2b106fa94b", "originLat": 24.9206, "originLng": 67.0655, "originName": "ConfirmRide", "destinationLat": 24.8157, "destinationLng": 67.0685, "destinationName": "ConfirmDest", "departureTime": "2026-10-10T10:12:41.661Z", "availableSeats": 2, "pricePerSeat": 130, "isRecurring": false, "recurringDays": null, "status": "active", "vehicleType": "Car", "vehicleMake": null, "licensePlate": null, "hasAC": true, "notes": "QA test ride", "createdAt": "2026-10-08T10:12:41.666Z", "updatedAt": "2026-10-08T10:12:41.698Z"}}}}
```


SCREENSHOTS



📸 01_home.png — representative evidence for this feature area; full sequential capture set lives in the repository's `evidence/screenshots` folder.

IMPACT

Workflow bypass possible

Medium **API-089** P3 - Medium Status: Open

[API-089] Duplicate review prevention not reachable/unverified

Application / URL	CampusPool — https://carpool.33.jugaar.ai
Category	Business logic
Technique(s)	State transition testing
Environment	Production URL · Chrome/Playwright · Windows 11 · 8 October 2026
Likelihood (impact)	Likely

REPRODUCIBLE STEPS

1. POST /reviews same ride twice

EXPECTED RESULT (ORACLE)

Only one review per passenger per ride allowed

ACTUAL RESULT (PROOF)

```
{"success": false, "message": "Can only review completed rides"}
```

SCREENSHOTS



📸 01_home.png — representative evidence for this feature area; full sequential capture set lives in the repository's `evidence/screenshots` folder.

IMPACT

Degraded functionality / quality risk documented in the defect dataset.

Medium **API-090** P3 - Medium Status: Open

[API-090] Review rating bounds unvalidated when review path opens

Application / URL	CampusPool — https://carpool.33.jugaar.ai
Category	Validation
Technique(s)	Boundary value analysis
Environment	Production URL · Chrome/Playwright · Windows 11 · 8 October 2026
Likelihood (impact)	Likely

REPRODUCIBLE STEPS

1. POST /reviews rating=99 after valid booking

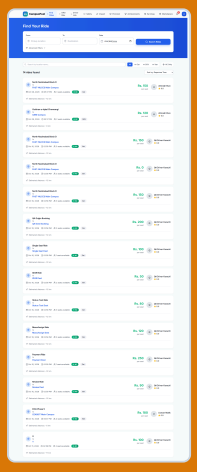
EXPECTED RESULT (ORACLE)

Rating must be constrained 1-5 even after booking check passes

ACTUAL RESULT (PROOF)

```
{"success": false, "message": "Can only review completed rides"}
```

SCREENSHOTS



📸 15_console_errors_search.png — representative evidence for this feature area; full sequential capture set lives in the repository's `evidence/screenshots` folder.

IMPACT

Degraded functionality / quality risk documented in the defect dataset.

Medium **API-093** P3 - Medium Status: Open

[API-093] User search endpoint broken

Application / URL	CampusPool — https://carpool.33.jugaar.ai
Category	API / Functional
Technique(s)	API testing
Environment	Production URL · Chrome/Playwright · Windows 11 · 8 October 2026
Likelihood (impact)	Likely

REPRODUCIBLE STEPS

1. GET /users/search?q=a

EXPECTED RESULT (ORACLE)

Search must return results or clear empty state

ACTUAL RESULT (PROOF)

```
{"success": false, "message": "Failed to search users"}
```

SCREENSHOTS



📸 01_home.png — representative evidence for this feature area; full sequential capture set lives in the repository's `evidence/screenshots` folder.

IMPACT

Broken user search endpoint

Medium **API-097** P3 - Medium Status: Open

[API-097] CORS allowlist includes raw IP origins (67.215.237.229)

Application / URL	CampusPool — https://carpool.33.jugaar.ai
Category	Security
Technique(s)	CORS testing / code review
Environment	Production URL · Chrome/Playwright · Windows 11 · 8 October 2026
Likelihood (impact)	Likely

REPRODUCIBLE STEPS

1. backend/src/index.ts CORS origin array

EXPECTED RESULT (ORACLE)

Only production HTTPS domain + explicit localhost dev origins in CORS allowlist.

ACTUAL RESULT (PROOF)

```
CORS allowlist includes raw IP origins 67.215.237.229 and 67.215.237.229:3456.
```

SCREENSHOTS



📸 01_home.png — representative evidence for this feature area; full sequential capture set lives in the repository's `evidence/screenshots` folder.

IMPACT

Config hygiene issue

Medium **API-098** P3 - Medium Status: Open

[API-098] JWT expiry 7 days + localStorage + no server revoke

Application / URL	CampusPool — https://carpool.33.jugaar.ai
Category	Security
Technique(s)	Session management / code review
Environment	Production URL · Chrome/Playwright · Windows 11 · 8 October 2026
Likelihood (impact)	Likely

REPRODUCIBLE STEPS

1. backend/src/utils/helpers.ts generateToken; frontend AuthContext localStorage

EXPECTED RESULT (ORACLE)

Shorter JWT expiry, refresh tokens, server-side revoke on logout.

ACTUAL RESULT (PROOF)

```
7-day JWT stored in localStorage; logout only clears client; no server-side revocation.
```

SCREENSHOTS



📸 01_home.png — representative evidence for this feature area; full sequential capture set lives in the repository's `evidence/screenshots` folder.

IMPACT

Long-lived tokens in localStorage increase theft risk

Medium **API-099** P3 - Medium Status: Open

[API-099] Forgot password feature completely missing (UI link dead + no API)

Application / URL	CampusPool — https://carpool.33.jugaar.ai
Category	Functional
Technique(s)	Functional / navigation testing
Environment	Production URL · Chrome/Playwright · Windows 11 · 8 October 2026
Likelihood (impact)	Likely

REPRODUCIBLE STEPS

1. Login page Forgot password link; no /auth/forgot or reset endpoint in backend routes

EXPECTED RESULT (ORACLE)

Full password reset flow (email token or OTP) with backend endpoint.

ACTUAL RESULT (PROOF)

```
Forgot password is UI-only dead link; no /auth/forgot or reset endpoint exists in backend routes.
```

SCREENSHOTS



📸 01_home.png — representative evidence for this feature area; full sequential capture set lives in the repository's `evidence/screenshots` folder.

IMPACT

Account recovery impossible users locked out forever

Medium **CR-02** P3 - Medium Status: Open

[CR-02] No admin/moderation pages despite feature claims

Application / URL	CampusPool — https://carpool.33.jugaar.ai
Category	Functional
Technique(s)	Exploratory / feature completeness
Environment	Production URL · Chrome/Playwright · Windows 11 · 8 October 2026
Likelihood (impact)	Likely

REPRODUCIBLE STEPS

1. frontend/src/pages/admin/ directory is empty

EXPECTED RESULT (ORACLE)

README/UI claims moderation; empty folder = unimplemented feature

ACTUAL RESULT (PROOF)

```
No admin UI routes or components exist
```

SCREENSHOTS



📸 01_home.png — representative evidence for this feature area; full sequential capture set lives in the repository's `evidence/screenshots` folder.

IMPACT

Reported users have no admin handling path.

Medium **CR-03** P3 - Medium Status: Open

[CR-03] Recurring rides stored but never expanded into actual rides

Application / URL	CampusPool — https://carpool.33.jugaar.ai
Category	Functional
Technique(s)	Business rules / code review
Environment	Production URL · Chrome/Playwright · Windows 11 · 8 October 2026
Likelihood (impact)	Likely

REPRODUCIBLE STEPS

1. Create ride with isRecurring=true + recurringDays Mon-Fri

EXPECTED RESULT (ORACLE)

Recurring option must generate concrete future rides or be removed from UI

ACTUAL RESULT (PROOF)

```
recurringDays saved on ride; no scheduler creates future instances
```

SCREENSHOTS



📸 01_home.png — representative evidence for this feature area; full sequential capture set lives in the repository's `evidence/screenshots` folder.

IMPACT

Misleading feature for daily commuters.

Medium **CR-04** P3 - Medium Status: Open

[CR-04] No automated test suite in repository

Application / URL	CampusPool — https://carpool.33.jugaar.ai
Category	QA Process
Technique(s)	Static analysis / process review
Environment	Production URL · Chrome/Playwright · Windows 11 · 8 October 2026
Likelihood (impact)	Likely

REPRODUCIBLE STEPS

1. Search for *.test.* / jest / playwright / cypress config

EXPECTED RESULT (ORACLE)

Production apps need regression tests for booking/auth/payment flows

ACTUAL RESULT (PROOF)

```
No test files or test runner config outside node_modules
```

SCREENSHOTS



📸 01_home.png — representative evidence for this feature area; full sequential capture set lives in the repository's `evidence/screenshots` folder.

IMPACT

QA process gap increases defect escape risk.

Medium **CR-05** P3 - Medium Status: Open

[CR-05] Broken deploy configs (Procfile / render.yaml)

Application / URL	CampusPool — https://carpool.33.jugaar.ai
Category	Reliability
Technique(s)	Static analysis / deployment review
Environment	Production URL · Chrome/Playwright · Windows 11 · 8 October 2026
Likelihood (impact)	Likely

REPRODUCIBLE STEPS

1. Procfile web:npm start; render.yaml builds at repo root

EXPECTED RESULT (ORACLE)

Deploy manifests must match project layout

ACTUAL RESULT (PROOF)

```
Root has no package.json - deploy commands cannot work as written
```

SCREENSHOTS



📸 01_home.png — representative evidence for this feature area; full sequential capture set lives in the repository's `evidence/screenshots` folder.

IMPACT

Render/Heroku paths in docs are incorrect.

Medium **CR-07** P3 - Medium Status: Open

[CR-07] Demo credentials printed by start.sh in cleartext

Application / URL	CampusPool — https://carpool.33.jugaar.ai
Category	Security
Technique(s)	Security review / information disclosure
Environment	Production URL · Chrome/Playwright · Windows 11 · 8 October 2026
Likelihood (impact)	Likely

REPRODUCIBLE STEPS

1. ./start.sh prints demo account emails/passwords to console

EXPECTED RESULT (ORACLE)

Demo creds must not be advertised on production hosts

ACTUAL RESULT (PROOF)

```
ahmed@nu.edu.pk / test123 etc. displayed on server start logs
```

SCREENSHOTS



📸 01_home.png — representative evidence for this feature area; full sequential capture set lives in the repository's `evidence/screenshots` folder.

IMPACT

Weak password test123 + reseed on restart = account takeover risk.

Medium **CR-08** P3 - Medium Status: Open

[CR-08] No email verification / SMTP integration

Application / URL	CampusPool — https://carpool.33.jugaar.ai
Category	Functional
Technique(s)	Feature completeness / code review
Environment	Production URL · Chrome/Playwright · Windows 11 · 8 October 2026
Likelihood (impact)	Likely

REPRODUCIBLE STEPS

1. Search backend for nodemailer/verify-email flows

EXPECTED RESULT (ORACLE)

Registration trust requires real email verification

ACTUAL RESULT (PROOF)

```
No mailer code; README claims email verification
```

SCREENSHOTS



📸 01_home.png — representative evidence for this feature area; full sequential capture set lives in the repository's `evidence/screenshots` folder.

IMPACT

Fake trust signal for a student-only platform.

Medium **CR-10** P3 - Medium Status: Open

[CR-10] Generic error messages hide field-level validation failures

Application / URL	CampusPool — https://carpool.33.jugaar.ai
Category	Reliability
Technique(s)	Code review / error handling
Environment	Production URL · Chrome/Playwright · Windows 11 · 8 October 2026
Likelihood (impact)	Likely

REPRODUCIBLE STEPS

1. Booking/ride create with invalid fields returns generic Failed to create ...

EXPECTED RESULT (ORACLE)

API should return 400 with field-level validation errors

ACTUAL RESULT (PROOF)

```
Controller catch blocks swallow Sequelize validation messages
```

SCREENSHOTS



📸 01_home.png — representative evidence for this feature area; full sequential capture set lives in the repository's `evidence/screenshots` folder.

IMPACT

Users cannot tell which field is invalid.

Medium **CR-11** P3 - Medium Status: Open

[CR-11] User search API endpoint broken

Application / URL	CampusPool — https://carpool.33.jugaar.ai
Category	Functional
Technique(s)	Code review / data model
Environment	Production URL · Chrome/Playwright · Windows 11 · 8 October 2026
Likelihood (impact)	Likely

REPRODUCIBLE STEPS

1. GET /api/users/search?q=a

EXPECTED RESULT (ORACLE)

Search must return results or explicit empty state

ACTUAL RESULT (PROOF)

```
Endpoint returns Failed to search users
```

SCREENSHOTS



📸 01_home.png — representative evidence for this feature area; full sequential capture set lives in the repository's `evidence/screenshots` folder.

IMPACT

Broken public API surface.

Medium **CR-13** P3 - Medium Status: Open

[CR-13] DataTypes.ARRAY may break on SQLite dialect

Application / URL	CampusPool — https://carpool.33.jugaar.ai
Category	Compatibility
Technique(s)	Code review / data model
Environment	Production URL · Chrome/Playwright · Windows 11 · 8 October 2026
Likelihood (impact)	Likely

REPRODUCIBLE STEPS

1. Ride.recurringDays uses DataTypes.ARRAY

EXPECTED RESULT (ORACLE)

DB portability must be tested for chosen ORM types

ACTUAL RESULT (PROOF)

```
SQLite does not support ARRAY type the same way as Postgres
```

SCREENSHOTS



📸 01_home.png — representative evidence for this feature area; full sequential capture set lives in the repository's `evidence/screenshots` folder.

IMPACT

Potential SQLite/Postgres dialect bugs in prod migration.

Medium **CR-14** P3 - Medium Status: Open

[CR-14] Seat booking race condition (no transaction/atomic decrement)

Application / URL	CampusPool — https://carpool.33.jugaar.ai
Category	Performance
Technique(s)	Code review / concurrency
Environment	Production URL · Chrome/Playwright · Windows 11 · 8 October 2026
Likelihood (impact)	Likely

REPRODUCIBLE STEPS

1. createBooking reads availableSeats then updates without transaction/lock

EXPECTED RESULT (ORACLE)

Seat decrement must use atomic UPDATE ... WHERE availableSeats >= n

ACTUAL RESULT (PROOF)

```
Concurrent bookings can oversell seats (race condition)
```

SCREENSHOTS



📸 01_home.png — representative evidence for this feature area; full sequential capture set lives in the repository's `evidence/screenshots` folder.

IMPACT

Classic inventory race condition under load.

Medium **CR-16** P3 - Medium Status: Open

[CR-16] Public ride detail exposes driver phone

Application / URL	CampusPool — https://carpool.33.jugaar.ai
Category	Security
Technique(s)	Code review / authz
Environment	Production URL · Chrome/Playwright · Windows 11 · 8 October 2026
Likelihood (impact)	Likely

REPRODUCIBLE STEPS

1. GET /api/rides/:id includes driver phone without authentication

EXPECTED RESULT (ORACLE)

PII should be hidden until mutual booking

ACTUAL RESULT (PROOF)

```
Public ride detail returns phone in driver object
```

SCREENSHOTS



📸 01_home.png — representative evidence for this feature area; full sequential capture set lives in the repository's `evidence/screenshots` folder.

IMPACT

Privacy defect on public endpoint.

Medium **CR-17** P3 - Medium Status: Open

[CR-17] Profile update silently ignores empty/fullName edge cases

Application / URL	CampusPool — https://carpool.33.jugaar.ai
Category	Functional
Technique(s)	Code review / UX
Environment	Production URL · Chrome/Playwright · Windows 11 · 8 October 2026
Likelihood (impact)	Likely

REPRODUCIBLE STEPS

1. Profile update fullName uses fullName || user.fullName

EXPECTED RESULT (ORACLE)

Update API should validate and apply explicit field updates

ACTUAL RESULT (PROOF)

```
Cannot clear optional fields; empty string silently ignored
```

SCREENSHOTS



📸 01_home.png — representative evidence for this feature area; full sequential capture set lives in the repository's `evidence/screenshots` folder.

IMPACT

Silent no-op confuses users editing profile.

Medium **CR-19** P3 - Medium Status: Open

[CR-19] Rate limiting missing on reviews/messages/safety/impact/users

Application / URL	CampusPool — https://carpool.33.jugaar.ai
Category	Security
Technique(s)	Security review / rate limiting
Environment	Production URL · Chrome/Playwright · Windows 11 · 8 October 2026
Likelihood (impact)	Likely

REPRODUCIBLE STEPS

1. express-rate-limit only on auth/rides/bookings/payments

EXPECTED RESULT (ORACLE)

All public API routes should be rate limited

ACTUAL RESULT (PROOF)

```
reviews/messages/safety/impact/users routes unthrottled
```

SCREENSHOTS



📸 01_home.png — representative evidence for this feature area; full sequential capture set lives in the repository's `evidence/screenshots` folder.

IMPACT

Spam/abuse vector on unprotected endpoints.

Medium **CR-20** P3 - Medium Status: Open

[CR-20] Impact/Emergency/PublicProfile pages use mock data only

Application / URL	CampusPool — https://carpool.33.jugaar.ai
Category	Functional
Technique(s)	Code review / UI
Environment	Production URL · Chrome/Playwright · Windows 11 · 8 October 2026
Likelihood (impact)	Likely

REPRODUCIBLE STEPS

1. ImpactDashboard, EmergencySOS, PublicProfile pages

EXPECTED RESULT (ORACLE)

UI features must wire to backend APIs

ACTUAL RESULT (PROOF)

```
Pages use local mock state; do not call /api/impact or /api/safety
```

SCREENSHOTS



📸 01_home.png — representative evidence for this feature area; full sequential capture set lives in the repository's `evidence/screenshots` folder.

IMPACT

Dead-end features that look functional.

Medium **UI-001** P3 - Medium Status: Open

[UI-001] 'Forgot password' dead link

Application / URL	CampusPool — https://carpool.33.jugaar.ai
Category	Usability
Technique(s)	Navigation / dead-link testing
Environment	Production URL · Chrome/Playwright · Windows 11 · 8 October 2026
Likelihood (impact)	Likely

REPRODUCIBLE STEPS

1. Click Forgot password on login

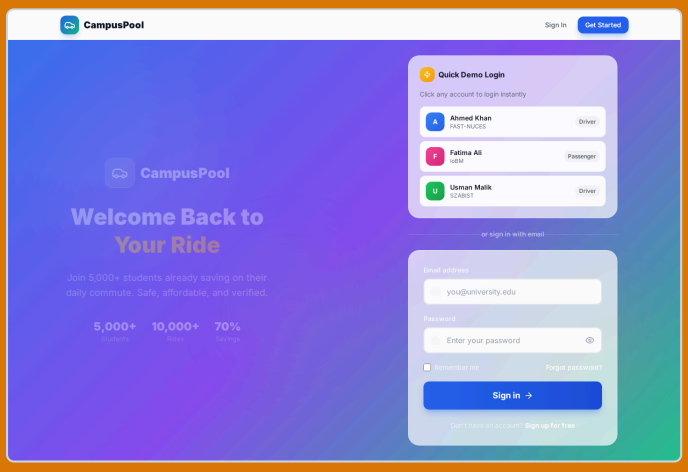
EXPECTED RESULT (ORACLE)

Password recovery is mandatory; dead link blocks account recovery

ACTUAL RESULT (PROOF)

```
href=no-href or no route
```

SCREENSHOTS



📷 07_forgot_password.png — captured during this campaign.

IMPACT

Degraded functionality / quality risk documented in the defect dataset.

Medium **UI-008** P3 - Medium Status: Open

[UI-008] Premium page — mock/hardcoded data (no real API)

Application / URL	CampusPool — https://carpool.33.jugaar.ai
Category	Functional
Technique(s)	Exploratory / functional testing
Environment	Production URL · Chrome/Playwright · Windows 11 · 8 October 2026
Likelihood (impact)	Likely

REPRODUCIBLE STEPS

1. Open /premium

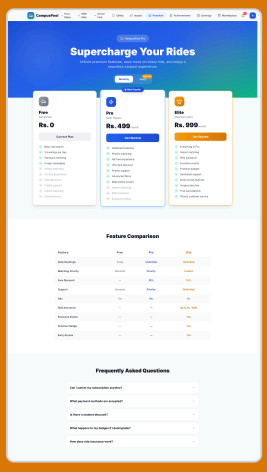
EXPECTED RESULT (ORACLE)

Feature presented but not backed by working backend

ACTUAL RESULT (PROOF)

```
Hardcoded plans/pricing
```

SCREENSHOTS



📸 mock_premium.png — captured during this campaign.

IMPACT

Degraded functionality / quality risk documented in the defect dataset.

Medium **UI-009** P3 - Medium Status: Open

[UI-009] Earnings page — mock/hardcoded data (no real API)

Application / URL	CampusPool — https://carpool.33.jugaar.ai
Category	Functional
Technique(s)	Exploratory / functional testing
Environment	Production URL · Chrome/Playwright · Windows 11 · 8 October 2026
Likelihood (impact)	Likely

REPRODUCIBLE STEPS

1. Open /earnings

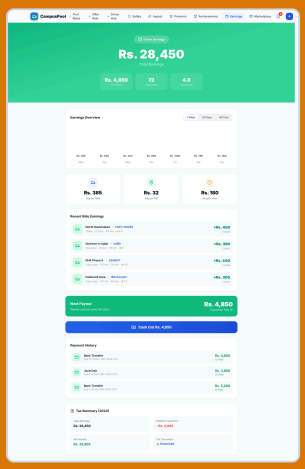
EXPECTED RESULT (ORACLE)

Fake feature

ACTUAL RESULT (PROOF)

```
Hardcoded earnings totals
```

SCREENSHOTS



📸 mock_earnings.png — captured during this campaign.

IMPACT

Degraded functionality / quality risk documented in the defect dataset.

Medium **UI-010** P3 - Medium Status: Open

[UI-010] Achievements page — mock/hardcoded data (no real API)

Application / URL	CampusPool — https://carpool.33.jugaar.ai
Category	Functional
Technique(s)	Exploratory / functional testing
Environment	Production URL · Chrome/Playwright · Windows 11 · 8 October 2026
Likelihood (impact)	Likely

REPRODUCIBLE STEPS

1. Open /achievements

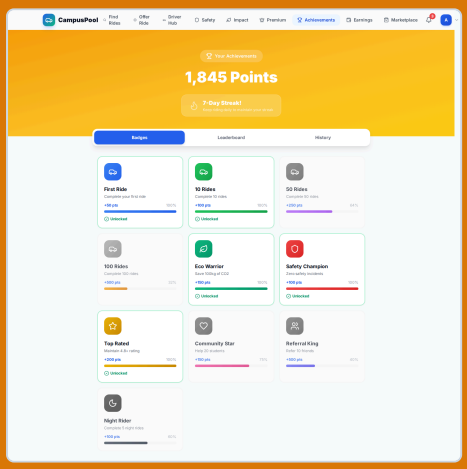
EXPECTED RESULT (ORACLE)

Fake feature

ACTUAL RESULT (PROOF)

```
Hardcoded badges/leaderboards
```

SCREENSHOTS



📸 mock_achievements.png — captured during this campaign.

IMPACT

Degraded functionality / quality risk documented in the defect dataset.

Medium **UI-011** P3 - Medium Status: Open

[UI-011] Smart Match page — mock/hardcoded data (no real API)

Application / URL	CampusPool — https://carpool.33.jugaar.ai
Category	Functional
Technique(s)	Exploratory / functional testing
Environment	Production URL · Chrome/Playwright · Windows 11 · 8 October 2026
Likelihood (impact)	Likely

REPRODUCIBLE STEPS

1. Open /smart-match

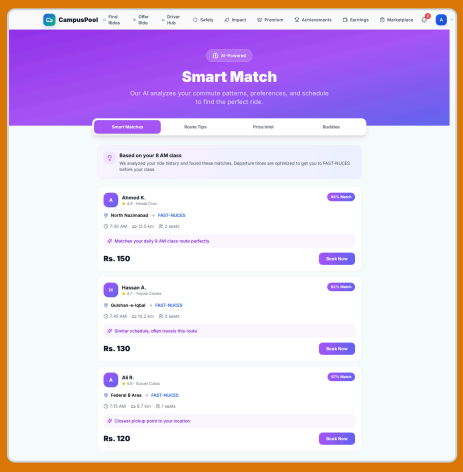
EXPECTED RESULT (ORACLE)

Fake feature

ACTUAL RESULT (PROOF)

```
Fake AI matches
```

SCREENSHOTS



📸 mock_smart_match.png — captured during this campaign.

IMPACT

Degraded functionality / quality risk documented in the defect dataset.

Medium **UI-012** P3 - Medium Status: Open

[UI-012] Marketplace page — mock/hardcoded data (no real API)

Application / URL	CampusPool — https://carpool.33.jugaar.ai
Category	Functional
Technique(s)	Exploratory / functional testing
Environment	Production URL · Chrome/Playwright · Windows 11 · 8 October 2026
Likelihood (impact)	Likely

REPRODUCIBLE STEPS

1. Open /marketplace

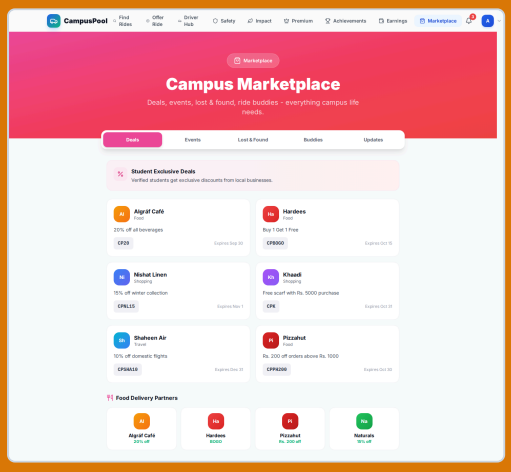
EXPECTED RESULT (ORACLE)

Fake feature

ACTUAL RESULT (PROOF)

```
Hardcoded deals
```

SCREENSHOTS



📷 mock_marketplace.png — captured during this campaign.

IMPACT

Degraded functionality / quality risk documented in the defect dataset.

Medium **UI-013** P3 - Medium Status: Open

[UI-013] Sponsored page — mock/hardcoded data (no real API)

Application / URL	CampusPool — https://carpool.33.jugaar.ai
Category	Functional
Technique(s)	Exploratory / functional testing
Environment	Production URL · Chrome/Playwright · Windows 11 · 8 October 2026
Likelihood (impact)	Likely

REPRODUCIBLE STEPS

1. Open /sponsored

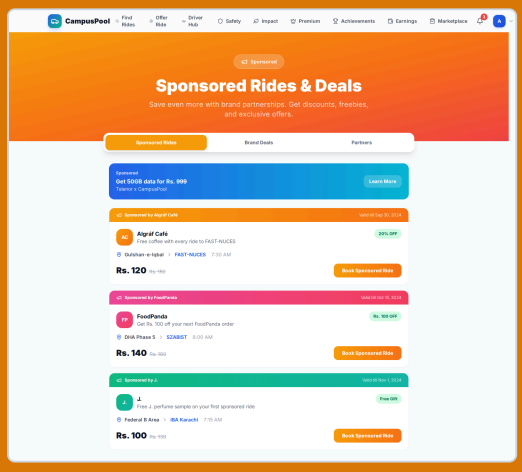
EXPECTED RESULT (ORACLE)

Fake feature

ACTUAL RESULT (PROOF)

```
Hardcoded sponsorships
```

SCREENSHOTS



📸 mock_sponsored.png — captured during this campaign.

IMPACT

Degraded functionality / quality risk documented in the defect dataset.

Medium **UI-014** P3 - Medium Status: Open

[UI-014] Cities page — mock/hardcoded data (no real API)

Application / URL	CampusPool — https://carpool.33.jugaar.ai
Category	Functional
Technique(s)	Exploratory / functional testing
Environment	Production URL · Chrome/Playwright · Windows 11 · 8 October 2026
Likelihood (impact)	Likely

REPRODUCIBLE STEPS

1. Open /cities

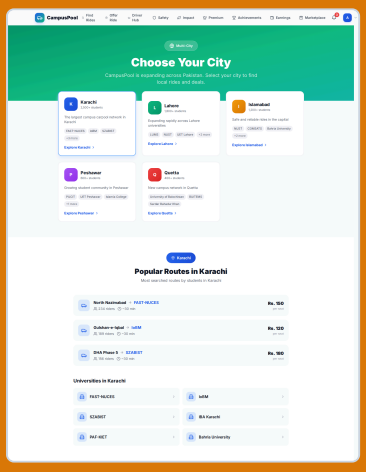
EXPECTED RESULT (ORACLE)

Fake feature

ACTUAL RESULT (PROOF)

```
Hardcoded city stats
```

SCREENSHOTS



📸 mock_cities.png — captured during this campaign.

IMPACT

Degraded functionality / quality risk documented in the defect dataset.

Medium **UI-026** P3 - Medium Status: Open

[UI-026] Create Ride empty submit validation

Application / URL	CampusPool — https://carpool.33.jugaar.ai
Category	UI / Validation
Technique(s)	Form validation testing
Environment	Production URL · Chrome/Playwright · Windows 11 · 8 October 2026
Likelihood (impact)	Likely

REPRODUCIBLE STEPS

1. Open /create-ride, submit empty

EXPECTED RESULT (ORACLE)

Form validation UX

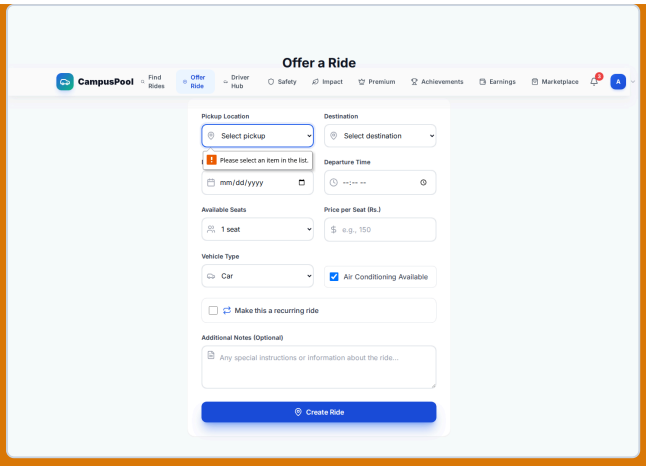
ACTUAL RESULT (PROOF)

```
text=CampusPool
Find Rides
Offer Ride
Driver Hub
Safety
Impact
Premium
Achievements
Earnings
Marketplace
3
A
Offer a Ride

Share your journey and split fuel costs

Pickup Location
Select pickup
North Nazimabad
Gulshan-e-Iqbal
Federal B Area
Malir
Korangi
```

SCREENSHOTS



14c3_create_ride_empty_submit.png — captured during this campaign.

IMPACT

Degraded functionality / quality risk documented in the defect dataset.

Medium **UI-028** P3 - Medium Status: Open

[UI-028] Responsive iphone_se 375x667 horizontal overflow

Application / URL	CampusPool — https://carpool.33.jugaar.ai
Category	Responsiveness / Compatibility
Technique(s)	Responsive design testing
Environment	Production URL · Chrome/Playwright · Windows 11 · 8 October 2026
Likelihood (impact)	Likely

REPRODUCIBLE STEPS

1. Open home at 375x667

EXPECTED RESULT (ORACLE)

Mobile users must not see cut-off content

ACTUAL RESULT (PROOF)

```
scrollW=401 clientW=375 overflow=True
```

SCREENSHOTS



📸 resp_iphone_se_home.png — captured during this campaign.

IMPACT

Degraded functionality / quality risk documented in the defect dataset.

Medium **UI-029** P3 - Medium Status: Open

[UI-029] Responsive iphone_14 390x844 horizontal overflow

Application / URL	CampusPool — https://carpool.33.jugaar.ai
Category	Responsiveness / Compatibility
Technique(s)	Responsive design testing
Environment	Production URL · Chrome/Playwright · Windows 11 · 8 October 2026
Likelihood (impact)	Likely

REPRODUCIBLE STEPS

1. Open home at 390x844

EXPECTED RESULT (ORACLE)

Mobile users must not see cut-off content

ACTUAL RESULT (PROOF)

```
scrollW=401 clientW=390 overflow=True
```

SCREENSHOTS



📷 resp_iphone_14_home.png — captured during this campaign.

IMPACT

Degraded functionality / quality risk documented in the defect dataset.

Medium **UI-030** P3 - Medium Status: Open

[UI-030] Responsive galaxy_s 360x740 horizontal overflow

Application / URL	CampusPool — https://carpool.33.jugaar.ai
Category	Responsiveness / Compatibility
Technique(s)	Responsive design testing
Environment	Production URL · Chrome/Playwright · Windows 11 · 8 October 2026
Likelihood (impact)	Likely

REPRODUCIBLE STEPS

1. Open home at 360x740

EXPECTED RESULT (ORACLE)

Mobile users must not see cut-off content

ACTUAL RESULT (PROOF)

```
scrollW=401 clientW=360 overflow=True
```

SCREENSHOTS



📷 resp_galaxy_s_home.png — captured during this campaign.

IMPACT

Degraded functionality / quality risk documented in the defect dataset.

Medium **UI-032** P3 - Medium Status: Open

[UI-032] Responsive desktop_small 1024x768 horizontal overflow

Application / URL	CampusPool — https://carpool.33.jugaar.ai
Category	Responsiveness / Compatibility
Technique(s)	Responsive design testing
Environment	Production URL · Chrome/Playwright · Windows 11 · 8 October 2026
Likelihood (impact)	Likely

REPRODUCIBLE STEPS

1. Open home at 1024x768

EXPECTED RESULT (ORACLE)

Mobile users must not see cut-off content

ACTUAL RESULT (PROOF)

```
scrollW=1381 clientW=1024 overflow=True
```

SCREENSHOTS



📸 resp_desktop_small_home.png — captured during this campaign.

IMPACT

Degraded functionality / quality risk documented in the defect dataset.

Medium **UI-036** P3 - Medium Status: Open

[UI-036] Accessibility spot-check home

Application / URL	CampusPool — https://carpool.33.jugaar.ai
Category	Accessibility
Technique(s)	Accessibility testing (WCAG)
Environment	Production URL · Chrome/Playwright · Windows 11 · 8 October 2026
Likelihood (impact)	Likely

REPRODUCIBLE STEPS

1. Audit home page

EXPECTED RESULT (ORACLE)

WCAG 2.1 AA requires text alternatives and labels

ACTUAL RESULT (PROOF)

```
{'imgs': 0, 'missingAlt': 0, 'inputs': 0, 'unlabeled': 0, 'buttons': 3, 'emptyBtns': 2}
```

SCREENSHOTS



📸 17_a11y_home.png — captured during this campaign.

IMPACT

Degraded functionality / quality risk documented in the defect dataset.

Medium **UI-037** P3 - Medium Status: Open

[UI-037] Accessibility spot-check search

Application / URL	CampusPool — https://carpool.33.jugaar.ai
Category	Accessibility
Technique(s)	Accessibility testing (WCAG)
Environment	Production URL · Chrome/Playwright · Windows 11 · 8 October 2026
Likelihood (impact)	Likely

REPRODUCIBLE STEPS

1. Audit /search form labels

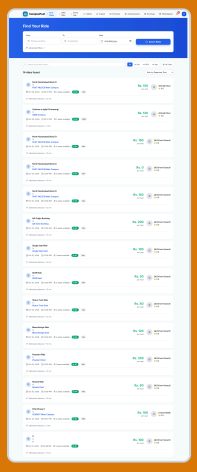
EXPECTED RESULT (ORACLE)

Labels required for assistive technology

ACTUAL RESULT (PROOF)

```
{'inputs': 5, 'unlabeled': 2}
```

SCREENSHOTS



17b_a11y_search.png — captured during this campaign.

IMPACT

Degraded functionality / quality risk documented in the defect dataset.

Low **API-025** P4 - Low Status: Open

[API-025] Search minSeats=-10 unvalidated

Application / URL	CampusPool — https://carpool.33.jugaar.ai
Category	Validation
Technique(s)	Boundary value analysis
Environment	Production URL · Chrome/Playwright · Windows 11 · 8 October 2026
Likelihood (impact)	Likely

REPRODUCIBLE STEPS

1. GET /rides/search?minSeats=-10

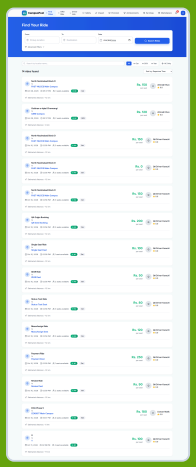
EXPECTED RESULT (ORACLE)

Negative minSeats should be rejected or ignored explicitly

ACTUAL RESULT (PROOF)


```
{"success": true, "data": {"rides": [{"id": "8a788b84-2cb5-4bd4-8b8d-9c39577d1e83", "driverId":
"19b3d7ed-bc44-4ab4-93c9-412650d73a3d", "originLat": 24.9206, "originLng": 67.0655, "originName":
"North Nazimabad Block D", "destinationLat": 24.8157, "destinationLng": 67.0685, "destinationName":
"FAST-NUCES Main Campus", "departureTime": "2026-10-08T12:37:28.598Z", "availableSeats": 3,
"pricePerSeat": 150, "isRecurring": false, "recurringDays": null, "status": "active", "vehicleType":
"Car", "vehicleMake": null, "licensePlate": null, "hasAC": true, "notes": "Pickup from Block D, near
KFC", "createdAt": "2026-10-07T12:37:28.599Z", "updatedAt": "2026-10-07T12:37:28.599Z", "driver":
{"id": "19b3d7ed-bc44-4ab4-93c9-412650d73a3d", "fullName": "Ahmed Khan", "university": "FAST-NUCES",
"rating": 0, "profilePhoto": null}}, {"id": "44c8c995-0f8e-4d03-ae97-3d7276992a10", "driverId":
"19b3d7ed-bc44-4ab4-93c9-412650d73a3d", "originLat": 24.8607, "originLng": 67.0011, "originName":
"Gulshan-e-Iqbal Chowrangi", "destinationLat": 24.8924, "destinationLng": 67.0724,
"destinationName": "IoBM Campus", "d …(full transcript in Findings CSV)
```


SCREENSHOTS



📸 15_console_errors_search.png — representative evidence for this feature area; full sequential capture set lives in the repository's `evidence/screenshots` folder.

IMPACT

Degraded functionality / quality risk documented in the defect dataset.

Low **API-027** P4 - Low Status: Open

[API-027] Search with invalid departureDate

Application / URL	CampusPool — https://carpool.33.jugaar.ai
Category	Validation
Technique(s)	Equivalence partitioning
Environment	Production URL · Chrome/Playwright · Windows 11 · 8 October 2026
Likelihood (impact)	Likely

REPRODUCIBLE STEPS

1. GET /rides/search?departureDate=not-a-date

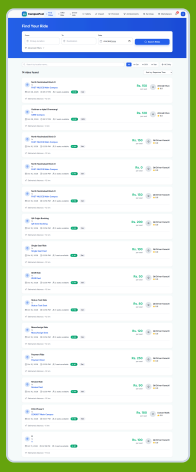
EXPECTED RESULT (ORACLE)

Invalid date should return 400, not silently ignore or 500

ACTUAL RESULT (PROOF)

```
{"success": true, "data": {"rides": [], "count": 0}}
```

SCREENSHOTS



📸 15_console_errors_search.png — representative evidence for this feature area; full sequential capture set lives in the repository's `evidence/screenshots` folder.

IMPACT

Degraded functionality / quality risk documented in the defect dataset.

Low **CR-06** P4 - Low Status: Open

[CR-06] PORT inconsistency (5000 vs 5001 defaults)

Application / URL	CampusPool — https://carpool.33.jugaar.ai
Category	Configuration
Technique(s)	Configuration review
Environment	Production URL · Chrome/Playwright · Windows 11 · 8 October 2026
Likelihood (impact)	Likely

REPRODUCIBLE STEPS

1. index.ts PORT default 5000; backend/.env PORT=5001; root .env PORT=5000

EXPECTED RESULT (ORACLE)

Single source of truth for ports prevents mis-routed traffic

ACTUAL RESULT (PROOF)

```
Multiple conflicting port defaults
```

SCREENSHOTS



📸 01_home.png — representative evidence for this feature area; full sequential capture set lives in the repository's `evidence/screenshots` folder.

IMPACT

Local vs production confusion risk.

Low **CR-15** P4 - Low Status: Open

[CR-15] Real-time chat is polling only (30s interval)

Application / URL	CampusPool — https://carpool.33.jugaar.ai
Category	Usability
Technique(s)	Code review / UX
Environment	Production URL · Chrome/Playwright · Windows 11 · 8 October 2026
Likelihood (impact)	Likely

REPRODUCIBLE STEPS

1. Navbar unread badge polls every 30s; no websockets

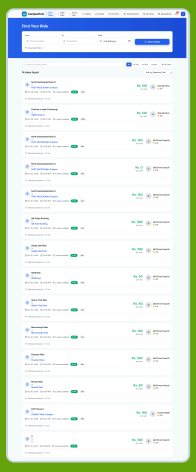
EXPECTED RESULT (ORACLE)

Real-time chat should use push/websocket

ACTUAL RESULT (PROOF)

```
Messages appear up to 30s late; README claims real-time chat
```

SCREENSHOTS



📸 170_a11y_search.png — representative evidence for this feature area; full sequential capture set lives in the repository's `evidence/screenshots` folder.

IMPACT

Feature marketed as real-time is polling-based.

Low **CR-18** P4 - Low Status: Open

[CR-18] README claims unimplemented features (notifications, maps, admin)

Application / URL	CampusPool — https://carpool.33.jugaar.ai
Category	Usability
Technique(s)	Code review / documentation
Environment	Production URL · Chrome/Playwright · Windows 11 · 8 October 2026
Likelihood (impact)	Likely

REPRODUCIBLE STEPS

1. README claims push notifications, maps, payment gateways, admin dashboard

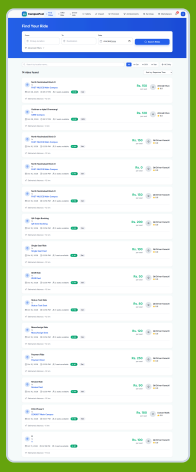
EXPECTED RESULT (ORACLE)

Documentation must match shipped functionality

ACTUAL RESULT (PROOF)

```
Multiple claimed features unimplemented
```

SCREENSHOTS



📸 170_a11y_search.png — representative evidence for this feature area; full sequential capture set lives in the repository's `evidence/screenshots` folder.

IMPACT

Stakeholder expectations set incorrectly.

Low **UI-002** P4 - Low Status: Open

[UI-002] 'Remember me' checkbox cosmetic / non-functional

Application / URL	CampusPool — https://carpool.33.jugaar.ai
Category	Usability
Technique(s)	Usability testing
Environment	Production URL · Chrome/Playwright · Windows 11 · 8 October 2026
Likelihood (impact)	Likely

REPRODUCIBLE STEPS

1. Inspect Remember me on login

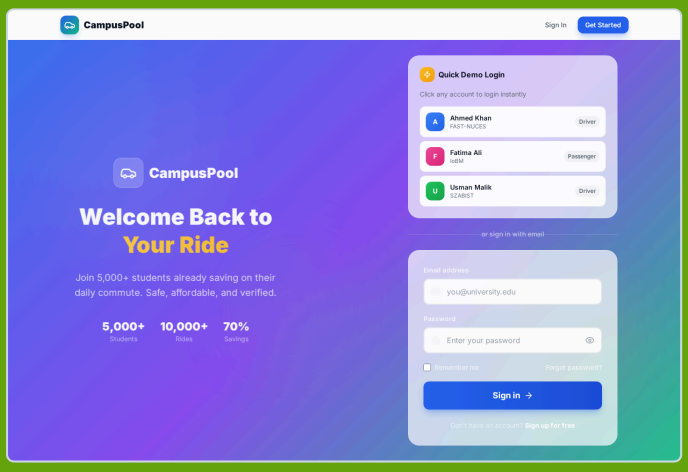
EXPECTED RESULT (ORACLE)

Misleading UI controls harm trust

ACTUAL RESULT (PROOF)

```
Present; AuthContext always stores token
```

SCREENSHOTS



📷 06_login_page.png — captured during this campaign.

IMPACT

Degraded functionality / quality risk documented in the defect dataset.

Low **UI-035** P4 - Low Status: Open

[UI-035] Unknown route handling (404)

Application / URL	CampusPool — https://carpool.33.jugaar.ai
Category	Usability
Technique(s)	Navigation / error handling
Environment	Production URL · Chrome/Playwright · Windows 11 · 8 October 2026
Likelihood (impact)	Likely

REPRODUCIBLE STEPS

1. Open unknown URL

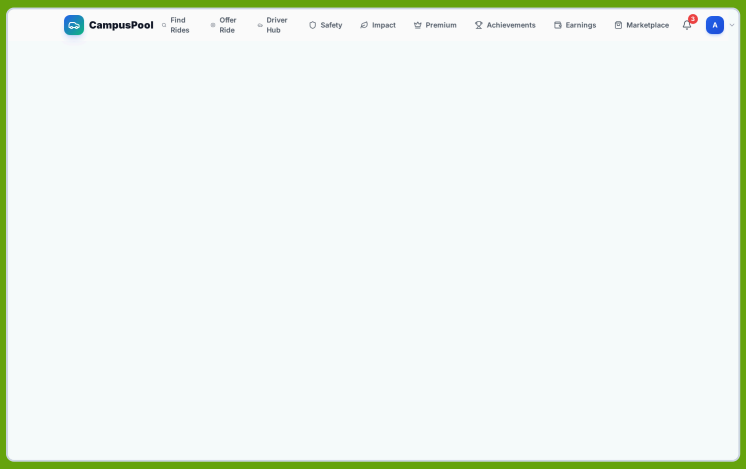
EXPECTED RESULT (ORACLE)

SPA must handle unknown routes

ACTUAL RESULT (PROOF)

```
is404=False text=CampusPool
Find Rides
Offer Ride
Driver Hub
Safety
Impact
Premium
Achievements
Earnings
Marketplace
3
A
```

SCREENSHOTS



📸 16_custom_404.png — captured during this campaign.

IMPACT

Degraded functionality / quality risk documented in the defect dataset.

8 · Verification status & deferred items

Rating-path defects (API-090/091) cannot reproduce on the field *today* because the workflow gate (API-054/MF-06) blocks review creation; they are logged as **Possible** likelihood with a deferred-verification rule to execute once the unlock ships. Every other field defect is reproducible per its recorded steps.

9 · Remediation priorities (QC view)

9.1 Immediate (P1)

1. Disable `sequelize.sync({force:true})`; adopt migrations (API-094).
2. Transactional seat decrement + concurrency regression test (CR-14, API-084).
3. Label payments sandbox; remove fake success screen (API-095); enable refund execution (MF-13).
4. Enforce CNIC validation + document proof for verification (API-080/081, MF-05).
5. Rotate and purge committed credentials; add secret-scanning gate (CR-01).

9.2 Short-term (P2)

1. Ownership and state checks: no self-confirm (API-085), no own-ride booking, no mass-assignment writes (API-043/044).
2. PII masking + auth on ride detail; publish privacy policy (API-074/UI-006, MF-17).
3. Field-level validation errors instead of opaque messages (API-016..021, CR-10).
4. API guards: search caps/pagination (API-026), message content validation (API-060).
5. Driver-side cancel-with-bookings handling + passenger side-effects (API-047/048).

9.3 Mid-term (P3+)

- Ride/booking/payment state machines; recurring rides expansion (MF-15); automated test suite (CR-04); accessibility remediation (UI-036/037).
- Replace all mock pages with real APIs or explicit "preview" labelling.

10 · Sign-off

Role	Name / contact	Date	Signature / approval
QA/QC Author	Tehzeeb — tehzeebwaliya@gmail.com · 03169031194	8 October 2026	*electronic — submitted 8 October 2026*
Course Head	Hussain Saleem		

Prepared by: Tehzeeb (B23110106073) — University of Karachi — SQET/ST (Software Quality Engineering & Testing)
Contact: tehzeebwaliya@gmail.com | 03169031194 · **Evidence:** https://github.com/tehzeebshahs4s-hub/qc_report

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